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## INTRODUCTORY ENGINEERING INTERNSHIP REPORT

Industrial Engineering Department

Industrial Engineering – Operational Excellence

First Year of the Engineering Cycle

# Digitalization and Design of an Intelligent Supervision Solution for the Soluble MAP Production Process

INDUSTRIAL AI · PROCESS ENGINEERING · DECISION SUPPORT

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Internship at OCP Group – Jorf Lasfar

**Internship Period:** 9 July–9 August 2026

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# Abstract

This introductory engineering internship addresses an Industrial Engineering problem at OCP Jorf Lasfar: the information used to assess final soluble monoammonium phosphate (MAP) moisture is delivered by laboratory analysis at an interval of approximately two hours, whereas process conditions evolve continuously. The work therefore investigates a digital supervision architecture focused on the drying stage, where final moisture is strongly influenced and product conformity can be affected rapidly.

The contribution is a decision-support **prototype** that combines a moisture soft sensor, unsupervised anomaly detection, explainable diagnostic logic and a centralized Power BI dashboard. Plant process variables are available continuously through the industrial monitoring environment. For the software demonstration, they are represented on a deliberately selected five-second replay and visualization grid, while laboratory density, product temperature and moisture remain structurally sparse at their approximately two-hour availability. Python performs preprocessing, residence-time-aware alignment, feature computation and inference; PostgreSQL stores idempotent process and model outputs and exposes dashboard-oriented views; Power BI uses DirectQuery to present process KPIs, predicted and measured moisture, anomaly risk, operational severity, probable subsystem, contributors and verification guidance.

The final demonstration uses one deterministic 92-day synthetic source because sufficient real historical and real-time industrial data were not available. The same chronology feeds exploration, feature engineering, moisture-model selection, anomaly modelling and a held-out TEST replay; the runtime loads the exact artifacts exported by Notebooks 03 and 04, including the governed 16-feature moisture contract. The prototype is grounded in the observed soluble MAP process and PCS7 context, but it is not connected to live PCS7, does not command equipment and has not received industrial-scale validation. Its principal result is therefore a tested technical feasibility demonstration and a structured path toward future validation on OCP data.

**Keywords:** Industrial Engineering; operational excellence; soluble MAP; soft sensor; anomaly detection; diagnosis; digital supervision; Power BI; PostgreSQL; decision support.

# Contents

|                                                                       |    |
|-----------------------------------------------------------------------|----|
| Acknowledgements                                                      | i  |
| Abstract                                                              | ii |
| List of Figures                                                       | iv |
| General Introduction                                                  | 1  |
| 1 Industrial Context and Internship Mission                           | 3  |
| 2 Soluble MAP Production Process                                      | 5  |
| 3 Project Context, Problem Statement and Objectives                   | 13 |
| 4 Data Acquisition, Preparation and Feature Engineering               | 15 |
| 5 AI Models: Moisture Prediction and Anomaly Diagnosis                | 22 |
| 6 Real-Time Prototype Architecture: Python, PostgreSQL and Simulation | 28 |
| 7 Power BI Supervision Dashboard                                      | 32 |
| 8 Results, Validation, Limitations and Future Work                    | 36 |
| General Conclusion                                                    | 40 |
| Bibliography                                                          | 41 |

# List of Figures

|            |                                                                                                                               |    |
|------------|-------------------------------------------------------------------------------------------------------------------------------|----|
| Figure 1.1 | Jorf Lasfar industrial context. . . . .                                                                                       | 3  |
| Figure 2.1 | Overall soluble MAP production flow, simplified for supervision analysis. . . . .                                             | 5  |
| Figure 2.2 | PCS7 neutralization screen (top) and detail crop centred on the three agitated vessels (bottom). . . . .                      | 7  |
| Figure 2.3 | PCS7 buffer-reservoir synoptic used to explain process decoupling and material delay. .                                       | 8  |
| Figure 2.4 | PCS7 evidence for the two crystallization effects (top) and centrifugal separation (bottom). .                                | 9  |
| Figure 2.5 | Drying-stage engineering zoom and the principal variables represented by the prototype. .                                     | 10 |
| Figure 2.6 | Selected on-site drying evidence (top) paired with the PCS7 drying supervision screen (bottom). . . . .                       | 11 |
| Figure 2.7 | Selected on-site evidence for downstream solids handling and product storage. . . . .                                         | 11 |
| Figure 2.8 | Process-to-data-to-AI chain used in the project. . . . .                                                                      | 12 |
| Figure 3.1 | Multi-rate timing problem: fast process stream versus sparse laboratory quality. . . . .                                      | 13 |
| Figure 3.2 | Real-time supervision architecture with explicit multi-rate timing, current/future sources and operator authority. . . . .    | 14 |
| Figure 4.1 | Distributions of process and sparse quality variables in the canonical source. . . . .                                        | 17 |
| Figure 4.2 | Feature ranges and potential outliers in the canonical source. . . . .                                                        | 17 |
| Figure 4.3 | Pearson correlation matrix for canonical process and sparse quality variables. . . . .                                        | 18 |
| Figure 4.4 | Selected aligned engineered-feature relationships with final moisture. . . . .                                                | 18 |
| Figure 4.5 | Machine-learning workflow embedded in the engineering development cycle. . . . .                                              | 21 |
| Figure 5.1 | Later-segment VALIDATION RMSE used to select Ridge after training-only tuning. . .                                            | 23 |
| Figure 5.2 | Actual and predicted moisture over the untouched chronological TEST interval. . . . .                                         | 24 |
| Figure 5.3 | Absolute standardized coefficients of the selected Ridge pipeline. . . . .                                                    | 25 |
| Figure 5.4 | Anomaly-to-diagnosis transformation implemented for real-time supervision. . . . .                                            | 26 |
| Figure 6.1 | Event-level workflow paced at the selected five-second prototype interval. . . . .                                            | 29 |
| Figure 7.1 | Final SQL-backed Operations Overview with separate data and process status, moisture supervision and process context. . . . . | 34 |
| Figure 7.2 | Final SQL-backed Diagnostics and Root Cause page with event context, ranked contributors and verification guidance. . . . .   | 34 |
| Figure 8.1 | Stage-gated path from the current prototype to a possible industrial advisory system. .                                       | 39 |

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# General Introduction

Industrial performance is not improved by isolated algorithms; it is improved when reliable information reaches the right decision-maker at the right time. In a continuous production setting, this means linking process understanding, quality management, information flow and operator action. This internship is consequently positioned as an **Industrial Engineering digitalization and supervision project using data and AI technologies**, rather than as an artificial-intelligence exercise detached from production.

At OCP Jorf Lasfar, the soluble monoammonium phosphate production chain comprises successive chemical and physical transformations, from reactant preparation and neutralization to crystallization, solid–liquid separation, drying and conditioning. The project narrows this chain to the drying stage because final moisture is strongly influenced there, moisture is a direct quality indicator, and abnormal drying behaviour can rapidly affect conformity. The industrial difficulty is temporal: key process measurements are continuously available to supervision, while laboratory quality results become available only about once every two hours. Operators therefore face a period in which the most recent laboratory result may no longer fully represent the current state.

The project examines whether this visibility gap can be reduced through a digital decision-support chain. A moisture soft sensor estimates final moisture from process behaviour; a novelty detector identifies operating observations that depart from a reference regime; and a diagnostic layer translates deviations into contributors, subsystems, cautious probable diagnoses and verification actions. A Python service replays prototype rows on a deliberately selected five-second cycle, PostgreSQL centralizes the data, and a Power BI DirectQuery report provides an operator-oriented interface. The five-second interval is a practical simulation and visualization frequency, not the native measurement frequency of the plant instrumentation.

The scope boundary is fundamental. The industrial process, observed PCS7 screens, laboratory workflow and selected variables are real anchors. The final development and demonstration datasets, however, are synthetic prototypes designed to reproduce realistic multi-rate behaviour. There is no direct connection to the live PCS7 system, no autonomous actuation, no closed-loop production control and no claim of industrial-scale validation. A five-second replay demonstrates the software architecture; it does not certify plant performance.

The adopted workflow follows the logic of an industrial improvement project: understand the process and the decision latency; establish a data contract; construct and audit the dataset; prevent temporal leakage; compare models; integrate inference with a traceable database; design a supervision interface; and validate the complete chain. Python, scikit-learn, PostgreSQL and Power BI are enabling tools within that workflow.

Three questions guide the study. First, can fast process history provide a useful estimate of a quality variable whose direct observation is sparse? Second, can abnormal operation be communicated in a form that supports physical verification rather than presenting an opaque model score? Third, can the complete information chain operate frequently enough for near-real-time supervision while preserving traceability and failure visibility? These questions connect modelling performance with the wider production system.

The project also adopts explicit information categories. A *measurement* is a value acquired from

the process or laboratory source. A *prediction* is a model output derived from a defined historical window. An *anomaly* is a statistical departure from the learned reference, not necessarily a fault. A *diagnosis* is a ranked, rule-supported interpretation, not proof of cause. A *recommendation* is a verification prompt, not an instruction to change a setpoint. Keeping these categories separate reduces automation bias and provides a basis for future operator training.

The expected operational benefit is consequently expressed in terms of visibility and response preparation, not as an unverified economic saving. A continuous estimate may reveal a developing quality tendency between lab results; an anomaly event may direct attention to the most relevant subsystem; and centralized history may shorten the time needed to reconstruct what changed. Quantified benefits such as reduced off-specification production, lower energy consumption or shorter response time would require baseline plant data and a controlled industrial evaluation that are outside the initiation-internship scope.

Finally, the report treats reproducibility as part of engineering quality. Dataset cadence and sample counts are audited, metrics are read from saved training records, model and feature-schema versions are persisted, database writes are idempotent, validation commands are recorded and the report figures are tied to repository evidence. This discipline makes the current limitations inspectable and prepares a better hand-off for a later industrial study.

This report is organized in eight chapters. Chapters 1 and 2 establish the concise industrial context and the soluble MAP process. Chapter 3 formalizes the problem, scope and objectives. Chapter 4 documents data construction, preparation and feature engineering. Chapter 5 presents the predictive and diagnostic models. Chapters 6 and 7 describe the prototype replay architecture and the Power BI dashboard. Chapter 8 consolidates results, tests, limitations and the future industrialization roadmap.

#### Interpretation of every result

All numerical model and runtime results in this report concern the repository's prototype/synthetic datasets. They demonstrate technical feasibility under controlled replay conditions, not production-grade accuracy, reliability, safety integrity or economic benefit.

## CHAPTER 1

# Industrial Context and Internship Mission

## 1.1 OCP and Jorf Lasfar in project context

OCP Group operates an integrated phosphate value chain spanning extraction, transformation and plant-nutrition products. For this project, the relevant context is the coordination of material, energy, equipment, quality and logistics rather than corporate history. Jorf Lasfar is an integrated industrial platform combining fertilizer and acid production with utilities, storage and port interfaces [2, 3]. Disturbances can therefore propagate across connected operations, making timely and traceable supervision an Industrial Engineering concern.



(a) Integrated processing platform.



(b) Port and logistics interface.

**Figure 1.1:** Jorf Lasfar industrial context.

*Source: Supplied ENSAM/OCP organization report [4]; contextual images, not evidence of prototype deployment.*

The internship focuses on the soluble monoammonium phosphate (MAP) installation. Its production sequence includes reactant preparation, neutralization, buffering, crystallization, centrifugal separation, drying, cooling and final handling. The supplied PCS7 screens confirm an established industrial monitoring environment across these stages. They informed process understanding, but the prototype has no live connection to PCS7 and does not reproduce its control logic.

## 1.2 Operational problem and digital relevance

Process variables are continuously available through the industrial control and monitoring environment, while the relevant laboratory quality results are available approximately every two hours. The last laboratory result is therefore an authoritative historical measurement, but it cannot by itself describe every change occurring before the next sample. The project addresses this information-latency gap by combining a moisture estimate, anomaly awareness and an operator-oriented diagnostic view.

Digitalization is useful here only when provenance remains visible. The dashboard must distinguish a process measurement, a laboratory measurement, a model prediction and a diagnostic hypothesis. It must also show freshness, model/schema version and unavailable states. These requirements prevent a carried-forward result or model output from being mistaken for a new plant measurement.



### 1.3 Internship mission and system boundary

The mission is to design and demonstrate an intelligent supervision architecture for soluble MAP production, with detailed implementation around dryer moisture and abnormal-operation visibility. Python handles data validation, time alignment, feature construction and inference; PostgreSQL provides traceable persistence; and Power BI brings process, quality and model evidence into one supervision interface.

The internship deliverable is a controlled proof of concept. It does not connect to live PCS7, write to equipment, automate a conformity decision, replace laboratory control or claim industrial-scale accuracy. Production personnel and approved OCP procedures retain authority. A future deployment would additionally require approved historian access, cybersecurity review, tag and clock validation, representative historical data, shadow-mode trials and jointly owned response procedures.

#### Industrial requirements carried into the design

Every displayed value must retain its provenance and timestamp; stale or incomplete data must remain visible; anomaly evidence must lead to a cautious verification prompt; model and feature versions must be traceable; and the analytical layer must remain outside the control loop.

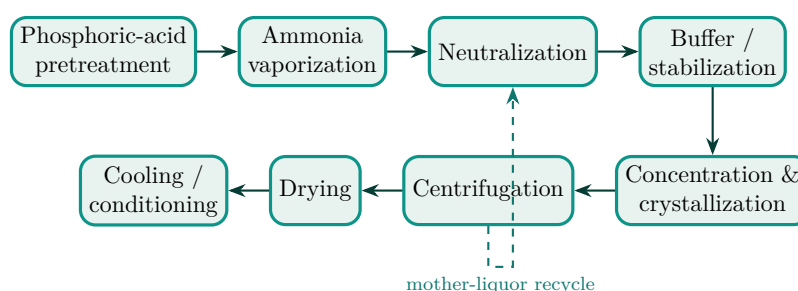
The next chapter develops the physical process basis for these requirements and explains why drying variables, material delay and sparse laboratory moisture must be interpreted together.

# Soluble MAP Production Process

## 2.1 From prepared reactants to conditioned crystals

This chapter reconstructs the soluble-MAP sequence from the supplied process report [5], the student's field observations [6], selected on-site photographs, and the project-specific MAP SOLUBLE PCS7 screens. The photographs establish physical context and the screenshots establish the supervised context. Neither is presented as evidence of a live prototype connection, and no equipment identity, PCS7 tag, unit, setpoint, alarm threshold or control function is inferred beyond what is visibly supported.

The simplified sequence in fig. 2.1 starts with phosphoric-acid pretreatment and ammonia vaporization. Neutralization forms ammonium phosphate solution, buffer reservoirs stabilize the downstream feed, concentration and vacuum crystallization form crystals, centrifugation removes mother liquor, and drying removes residual water before cooling and conditioning. Recycle streams and intermediate storage couple the stages, so a final-moisture deviation cannot automatically be attributed to one dryer component.



**Figure 2.1:** Overall soluble MAP production flow, simplified for supervision analysis.  
*Source: Author's synthesis from [5, 6] and the supplied MAP SOLUBLE PCS7 screens.*

### How to read the PCS7 evidence

The screen title identifies the process stage; equipment symbols and line connections support a qualitative description of material flow and supervision. Numerical values are point-in-time observations photographed from PCS7. The on-site photographs are selected for clarity and technical relevance, but visual similarity alone does not establish equipment identity or a one-to-one tag mapping. The report therefore connects photographs, PCS7 and the dataset only at confirmed engineering-concept level.

## 2.2 Reactant preparation and upstream environmental control

The cited process description presents phosphoric-acid pretreatment through desulfation, dearsenation, flocculation/decantation and aluminization [5]. These operations influence solution composition before neutralization but are not direct inputs to the current dryer model. The project-specific PCS7 and on-site evidence therefore receive visual priority.

Liquid ammonia is vaporized before neutralization. The supplied PCS7 views show vaporization and acid scrubbing as separate supervised utilities, establishing phase-change and gas-handling

context without justifying a tag-to-repository mapping. Both remain outside the model-input boundary.

### 2.3 Neutralization and stabilization

The principal reaction may be represented by



with further ammoniation capable of producing diammonium phosphate depending on ratio and conditions [5]. Neutralization is exothermic. Acid/ammonia balance, pH, temperature, agitation and recycled mother liquor therefore belong to one interacting operating context. These concepts are supported by the process description; the current prototype has no direct neutralization columns and does not claim to reconstruct the PCS7 control loops.

The full screen in fig. 2.2 is explicitly identified by its interface title as the neutralization stage. It shows three agitated vessel symbols labelled 207A-AR-201, 207A-AR-202 and 207A-AR-203, their associated drives/pumps, interconnected process lines and labelled connections for reagent, condensate, mother-liquor and gas services. The detail crop makes the vessel train readable without translating the surrounding tag codes or treating the displayed values as limits. None of the new on-site photographs provides a sufficiently reliable identifier for these three reactors; a physical pairing is therefore deliberately not asserted.

Buffer reservoirs absorb short-term variation and decouple neutralization from downstream processing. The screen in fig. 2.3 is explicitly identified by its interface title as the buffer-reservoir stage and visibly presents four agitated reservoirs with connected pumps and process lines. Its relevance is temporal: buffering delays and mixes disturbances, so an upstream change need not appear at the final product at the same timestamp.

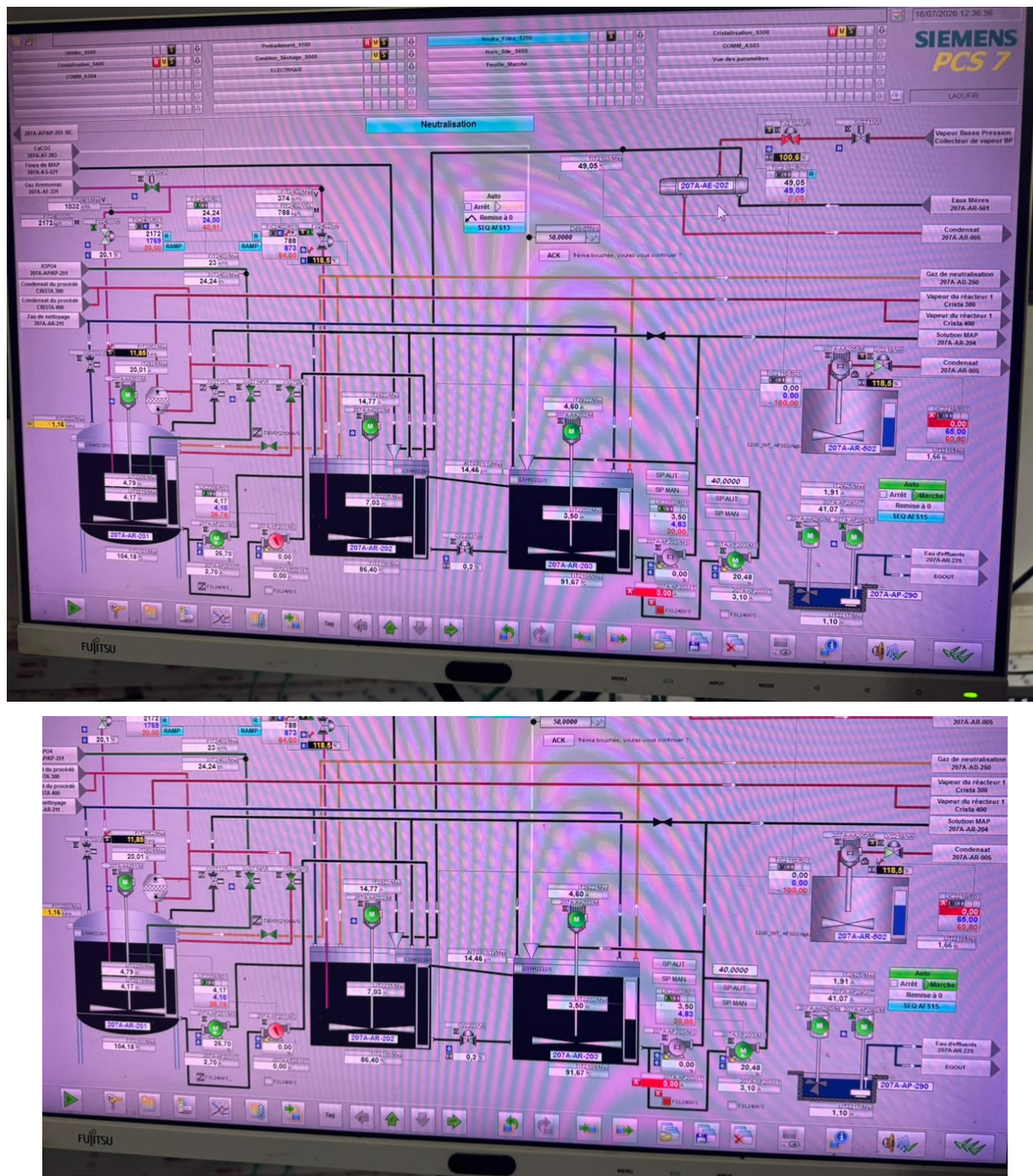
### 2.4 Crystallization and centrifugal separation

Crystallization requires supersaturation. In the observed process, circulation, vacuum operation and thermal exchange support concentration and crystal formation. The two supplied PCS7 views identify the 300 and 400 second-effect crystallization lines. Their side-by-side presentation in fig. 2.4 confirms that the operating context is distributed across two labelled effects/lines rather than represented by a single vessel. No displayed value is interpreted as an approved setpoint or industrial limit.

The suspension leaving crystallization is sent to centrifugal separation. The corresponding PCS7 screen visibly shows two centrifuge symbols, feed/service connections, motors and downstream collection. Separation removes much of the mother liquor before thermal drying. The engineering link is important even though no centrifuge tag is present in the dataset: wetter cake raises the evaporation duty imposed on the dryer, and upstream load should remain a diagnostic hypothesis rather than being mislabelled as a confirmed dryer failure.

### 2.5 Drying-stage process zoom

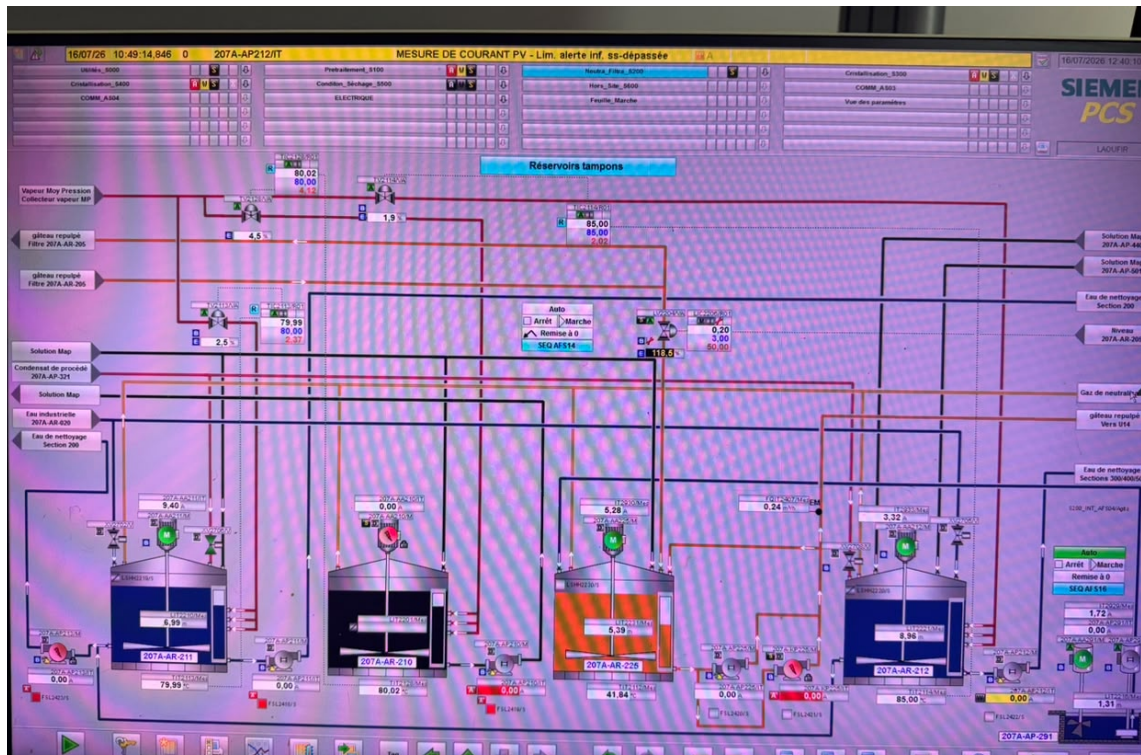
The dryer receives wet crystals whose residual liquid must be reduced before final handling. Hot drying air supplies heat and mass-transfer capacity, while product loading, gas-flow condition and exposure determine the achieved moisture. Cooling then reduces product temperature. Figure 2.5



**Figure 2.2:** PCS7 neutralization screen (top) and detail crop centred on the three agitated vessels (bottom).

Source: MAP SOLUBLE PCS7 photograph supplied by the author; tags and values reproduced only as visible interface evidence.





**Figure 2.3:** PCS7 buffer-reservoir synoptic used to explain process decoupling and material delay.  
*Source: MAP SOLUBLE PCS7 photograph supplied by the author; no tag-to-dataset mapping asserted.*

translates this physical interpretation into the variables defined by the repository.

Figure 2.6 connects three levels of evidence. Two selected on-site photographs show the drying-section structure and the rotary shell; the PCS7 view shows the supervised dryer train and downstream path. The physical form and the labelled screen support a functional drying-stage pairing, but no photographed component is assigned to a PCS7 tag. The repository data dictionary remains the authority for prototype variables and units.

The project defines nine fast process inputs: dryer-air temperature, cooler-air temperature, air-flow rate, wet-product feed rate, product-inlet temperature, residence time, vacuum, steam pressure and fan speed. Product density, final product temperature and final moisture are sparse two-hour quality variables in the demonstration contract [8]. Engineered balances such as air per feed, temperature differences and thermal exposure encode interactions without claiming to be first-principles energy meters.

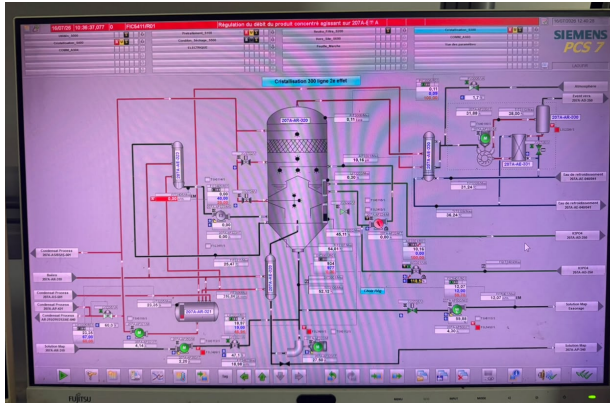
Downstream evidence is selected just as conservatively. Figure 2.7 shows one clear transfer installation and one storage view where soluble-MAP packaging is visibly identifiable. Repeated hopper close-ups, lower-quality transfer angles and packaging-film images that could not be tied specifically to soluble MAP were excluded.

### 2.5.1 Mass- and heat-transfer interpretation

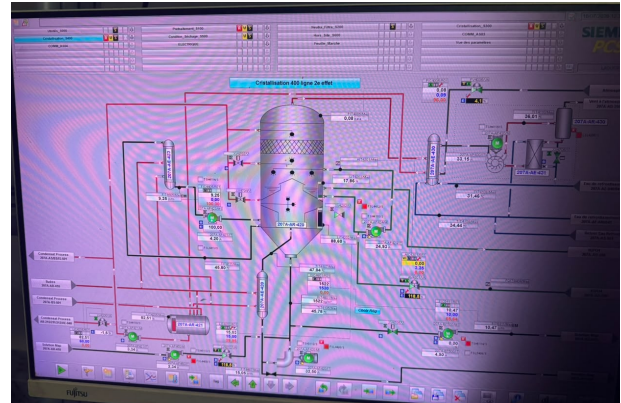
If  $\dot{m}_s$  is dry-solid flow and  $X_{in}$  and  $X_{out}$  are water-to-dry-solid ratios, the approximate evaporation demand is

$$\dot{m}_{evap} = \dot{m}_s (X_{in} - X_{out}). \quad (2.2)$$

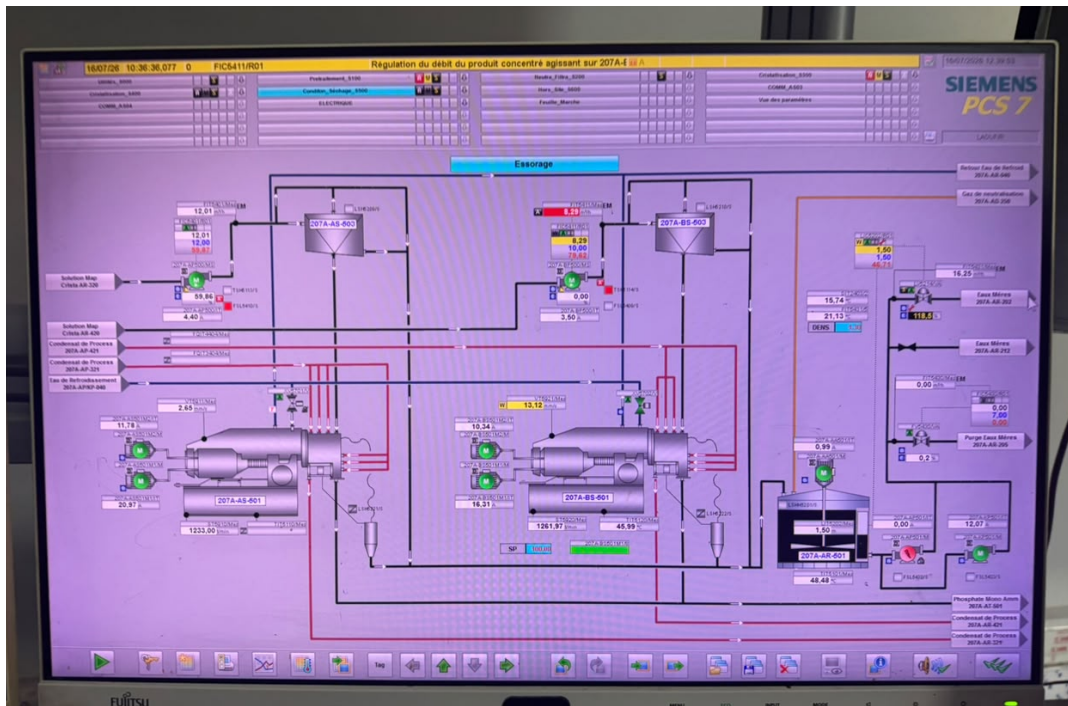
This balance is interpretive rather than an online calculation because wet-cake moisture and dry-solids flow are not present on every row of the prototype replay grid. It nevertheless explains



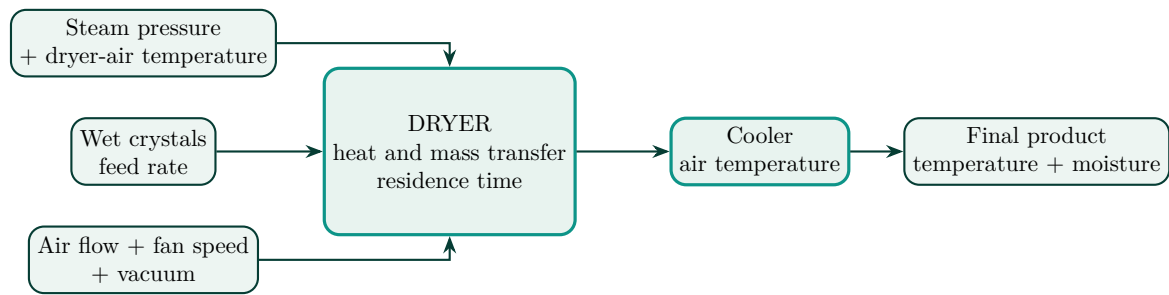
(a) Crystallization 300, second effect.



(b) Crystallization 400, second effect.



**Figure 2.4:** PCS7 evidence for the two crystallization effects (top) and centrifugal separation (bottom).  
*Source: MAP SOLUBLE PCS7 photographs supplied by the author; qualitative process evidence only.*



**Figure 2.5:** Drying-stage engineering zoom and the principal variables represented by the prototype.  
*Source: Author's synthesis from the repository data dictionary and feature metadata [8].*

why increased feed or wetter centrifuge cake raises demand, while insufficient heat or drying-air transfer can raise final moisture. The five-second grid used later is a deliberate simulation and visualization choice; it is not asserted as the native frequency of these plant measurements.

Steam pressure and dryer-air temperature act as prototype indicators of thermal supply; air flow, fan speed and vacuum describe the air/exhaust path; wet-product feed and residence time describe loading and exposure. Changing one variable is not an automatic recommendation. Higher heat or air may affect energy use, final temperature or other constraints, while feed reduction affects production. The model therefore produces evidence and verification prompts only; PCS7 and approved OCP procedures remain authoritative.

## 2.6 From the physical process to the analytical layer

The analytical boundary is concentrated on the dryer. Neutralization, buffering, crystallization and centrifugation explain upstream chemistry, delay and wet-cake loading, but no direct tag mapping from those PCS7 screens is claimed. The implemented dataset uses verified dryer concepts: thermal supply, drying air/exhaust, product loading and exposure, cooling, and sparse final-quality observations.

Thermal variables and air/feed conditions enter the moisture and anomaly paths; residence time also supports temporal alignment. Final moisture remains the sparse supervised target. Density and final product temperature enter the moisture model only as the most recent strictly previous laboratory results; they are not treated as five-second measurements. The diagram communicates the implemented chain without implying that every upstream stage is a model input or that a PCS7 screen tag equals a repository column.

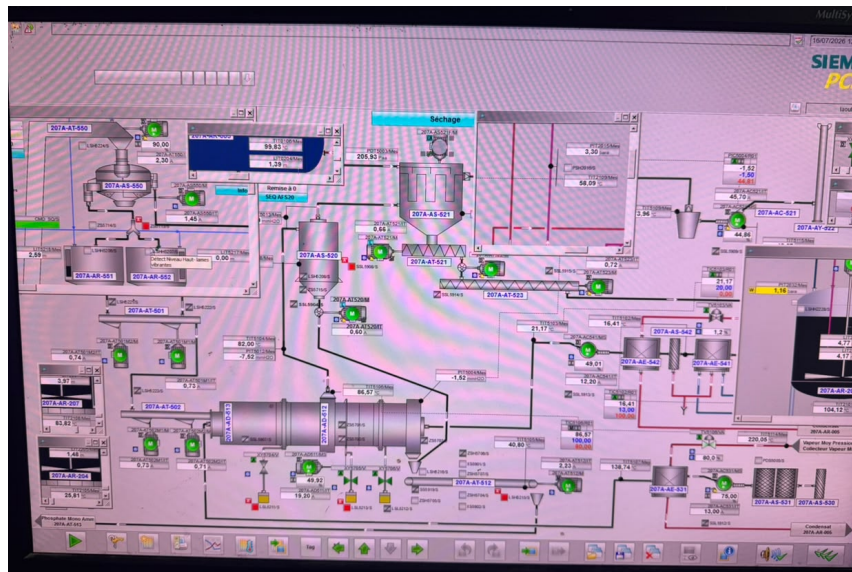




(a) On-site drying-section structure and process ductwork.



(b) On-site view of the rotary dryer shell.



**Figure 2.6:** Selected on-site drying evidence (top) paired with the PCS7 drying supervision screen (bottom).

*Source: On-site photographs and MAP SOLUBLE PCS7 screen supplied by the author; functional-stage pairing only, with no tag equivalence or live connection claimed.*



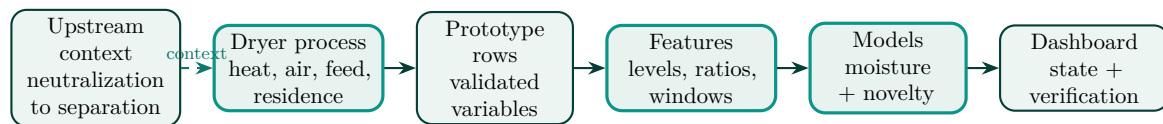
(a) On-site hopper and enclosed solids-transfer equipment; no equipment tag is asserted.



(b) Palletized soluble-MAP product in the finished-product storage area.

**Figure 2.7:** Selected on-site evidence for downstream solids handling and product storage.  
*Source: Original on-site photographs supplied by the author; aspect ratios preserved and technical content unaltered.*





**Figure 2.8:** Process-to-data-to-AI chain used in the project.

*Source: Author's synthesis from the repository data contract and implemented architecture [8, 7].*

## 2.7 Moisture supervision and temporal interpretation

Final moisture is selected because it is strongly influenced during drying, is directly linked to product quality, and can change before the next two-hour laboratory result. This does not imply a single cause. Thermal supply, drying-air delivery, exhaust condition, feed loading, upstream wetness, effective residence and cooling context can interact.

Process monitoring also involves delay. Neutralization and crystallization disturbances do not appear instantaneously at the final product because reservoirs, circulation, crystallizers, centrifuges and the dryer mix and delay material. The available repository does not track each parcel, so the supervised alignment explicitly accounts for dryer residence time and leaves additional transport delay configurable. This corrects the most direct temporal mismatch without claiming a complete dynamic model of the plant.

The PCS7 and Power BI roles are complementary. PCS7 is suited to equipment state and control-loop detail. The analytical layer combines longer time horizons, laboratory age, model validation and cross-variable evidence. When an event appears in Power BI, the operator returns to PCS7, laboratory evidence and approved field procedures for verification. The prototype neither writes to PCS7 nor changes a setpoint.

The laboratory closes the quality loop. A measured result is associated with sampling and analysis, while a prediction uses the residence-adjusted process snapshot and the strictly previous laboratory density and final-product-temperature results. The database preserves both values and evaluates error only at a true sample timestamp. If they disagree, the appropriate investigation covers sampling, time alignment, input quality, model support and process state; the prediction never overwrites the laboratory record.

### On-site and supervision evidence boundary

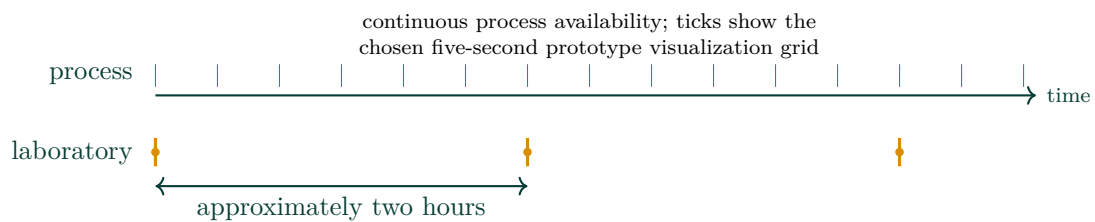
The selected on-site photographs document physical equipment and product handling, while the PCS7 photographs document the observed supervision context. Neither proves live prototype deployment. A photo-to-PCS7 relationship is asserted only at a verified functional stage; uncertain equipment identities and one-to-one tag mappings are explicitly withheld.

# Project Context, Problem Statement and Objectives

## 3.1 The decision-latency problem

The present monitoring context combines rich process visibility with delayed quality confirmation. PCS7 provides the industrial supervision environment, while laboratory analyses supply moisture, density and final product temperature approximately every two hours. Between two analyses, an operator can observe that temperatures, air flow, feed, vacuum or pressure have changed but cannot obtain a new measured moisture value immediately. The last laboratory result remains valid as a historical observation, not as proof of the current condition.

This is an Industrial Engineering problem of information latency and production decision support. The goal is to reduce the time needed to form a reasoned operational interpretation, while maintaining the distinction between measurement, prediction and diagnosis. The proposed solution inserts a digital information layer; it does not alter the process control layer.



**Figure 3.1:** Multi-rate timing problem: fast process stream versus sparse laboratory quality.  
*Source: Author's work from the MAP-Dryer-AI repository.*

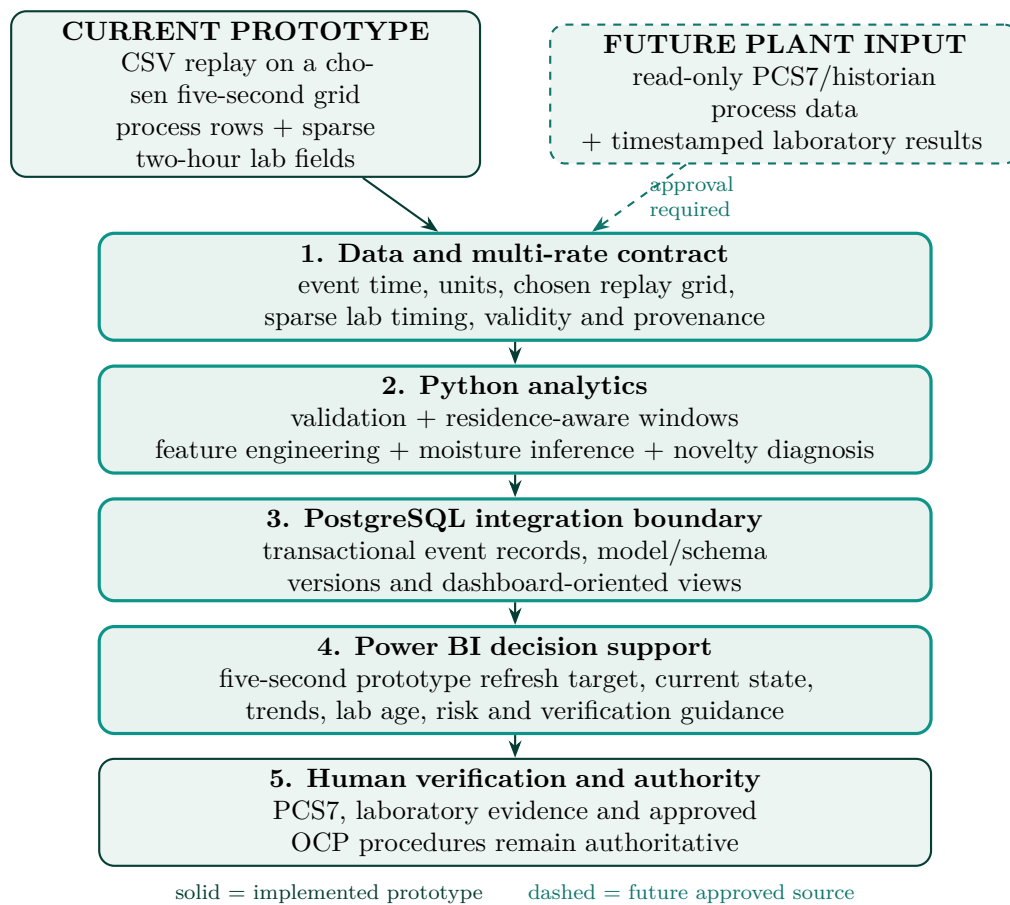
## 3.2 Project objectives and success criteria

The project objectives are to:

1. digitalize supervision of the soluble MAP process;
2. reduce dependence on delayed laboratory moisture availability through a continuous estimate;
3. detect abnormal operating conditions as early as possible;
4. identify the most likely affected subsystem when an anomaly occurs;
5. provide diagnostic information that supports faster operator decisions;
6. centralize process KPIs, model outputs, alerts and quality information in Power BI;
7. demonstrate a near-real-time architecture using Python, PostgreSQL and Power BI; and
8. create a prototype that could later receive live PCS7/historian data.

Success is assessed at prototype level through four evidence groups: correct multi-rate data handling; honest chronological model evaluation; complete replay on the chosen five-second demonstration grid with bounded computation time; and a structurally valid dashboard whose field references resolve to the semantic model. A successful prototype is not equivalent to a production acceptance test.

### 3.3 Proposed architecture and responsibility boundary



**Figure 3.2:** Real-time supervision architecture with explicit multi-rate timing, current/future sources and operator authority.

*Source: Author's work from the MAP-Dryer-AI repository.*

Python is responsible for data validation, temporal alignment, features, inference and diagnostic transformation. PostgreSQL provides idempotent persistence and stable analytical views. Power BI performs visualization, filtering and operator navigation; it does not run autonomous control logic. In the demonstration, the left-hand sources in fig. 3.2 are replaced by the held-out TEST interval of the same canonical CSV used by Notebooks 01–04, and the service loads the moisture artifact selected in Notebook 03 and the anomaly artifacts exported by Notebook 04. A future live interface must pass OCP cybersecurity, data-governance and operating-procedure requirements before the boundary can move.

#### Five-second interval: exact meaning

Process variables are available continuously through the industrial monitoring and control environment. The five-second interval was selected only for the prototype CSV grid, replay pacing and Power BI refresh demonstration so that near-real-time behaviour could be visualized consistently. It is not presented as the native acquisition or measurement frequency of the OCP instrumentation.

**Explicit non-objectives.** The project does not connect to live PCS7, actuate valves or drives, execute closed-loop control, certify a safety function, replace the laboratory, or claim production deployment. Its outputs are advisory and require operator verification.

# Data Acquisition, Preparation and Feature Engineering

## 4.1 Data strategy and contracts

Because a sufficiently large historical industrial dataset was not available, the final implementation uses the supplied deterministic canonical synthetic CSV for development and demonstration. The file spans 16 April–16 July 2026: exactly 92 days and 1,589,760 unique timestamps on a gap-free five-second grid. It carries 1,104 sparse laboratory observations at the configured two-hour cadence. Notebook 01 audits this source, Notebook 02 derives residence-aware supervised rows, Notebook 03 performs moisture-model selection and final evaluation, Notebook 04 exports the anomaly/diagnosis runtime artifacts, and the runtime replays only the held-out chronological TEST interval from the same source. The five-second interval is a prototype representation for replay, recurring inference and visualization, not evidence of native plant acquisition frequency. The supplied filename, dataset version and SHA-256 are recorded in the canonical manifest and propagated into the model artifacts [10, 7].

**Table 4.1:** Audited canonical data products used by the final implementation

| Dataset                    | Rows      | Columns | Lab rows | Cadence      | Role                                                                            |
|----------------------------|-----------|---------|----------|--------------|---------------------------------------------------------------------------------|
| Canonical source CSV       | 1,589,760 | 17      | 1,104    | 5 s; lab 2 h | Sole source for exploration, feature engineering, model development and replay. |
| Aligned laboratory handoff | 1,103     | 18      | 1,103    | 2 h          | Timestamp, 16 original named inputs and final-moisture target for Notebook 03.  |
| Held-out TEST replay       | 237,600   | 18      | 165      | 5 s; lab 2 h | Derived TEST interval used by the runtime, PostgreSQL and Power BI.             |

*Source: Repository CSV audit performed for this report.*

The raw prototype schema contains **Date**, **Time**, nine process variables and three sparse quality variables. Date and time are combined without timezone conversion because the project treats them as plant-local naive timestamps. The preprocessing module separates the single CSV into process and laboratory tables so that the absence of an unscheduled laboratory result is not mistaken for a missing sensor value.

The corresponding implementation keeps timestamp reconstruction and stream separation explicit:

**Listing 4.1:** Loading and separating the prototype process and laboratory streams.

```
raw = load_raw_source(source_path)
raw["Timestamp"] = build_timestamp(raw)

process, audit = build_process_table(
    raw, expected_interval_seconds=5,
    short_gap_fill_limit_intervals=0,
)
```

**Table 4.2:** Prototype variable contract and engineering units

| Variable                                           | Unit                           | Prototype representation | Engineering interpretation                                         |
|----------------------------------------------------|--------------------------------|--------------------------|--------------------------------------------------------------------|
| Dryer / cooler air temperature                     | °C                             | Each replay row          | Thermal state and cooling condition.                               |
| Air-flow rate                                      | m <sup>3</sup> h <sup>-1</sup> | Each replay row          | Drying-air delivery.                                               |
| Wet-product feed rate                              | m <sup>3</sup> h <sup>-1</sup> | Each replay row          | Dryer loading.                                                     |
| Product-inlet temperature                          | °C                             | Each replay row          | Incoming material condition.                                       |
| Residence time                                     | min                            | Each replay row          | Product exposure and alignment shift.                              |
| Vacuum                                             | mmH <sub>2</sub> O             | Each replay row          | Exhaust/depression state; more negative means stronger depression. |
| Steam pressure                                     | bar                            | Each replay row          | Thermal-energy supply proxy.                                       |
| Fan speed                                          | rpm                            | Each replay row          | Air/exhaust equipment state.                                       |
| Density; final product temperature; final moisture | kg/L; °C; % H <sub>2</sub> O   | approx. 2 h              | Laboratory reference; never forward-filled as new measurement.     |

Source: Data dictionary and preprocessing contract [8, 7].

```
laboratory = build_lab_table(raw)

assert audit.duplicate_timestamps_removed == 0
assert audit.missing_interval_rows == 0
```

The checks validate the chosen prototype grid; they do not infer the plant historian’s native sampling configuration. Laboratory rows are extracted from genuinely populated quality fields, while structural blanks remain blank.

## 4.2 Cleaning and exploratory analysis

The ingestion contract parses timestamps, coerces expected numeric columns, checks uniqueness and reports gaps rather than silently inventing rows. The audited canonical source contains no duplicates or missing steps on its selected grid, and every process field is populated. The 1,588,656 blank values per quality column are therefore *structural sparsity*, not cleaning defects.

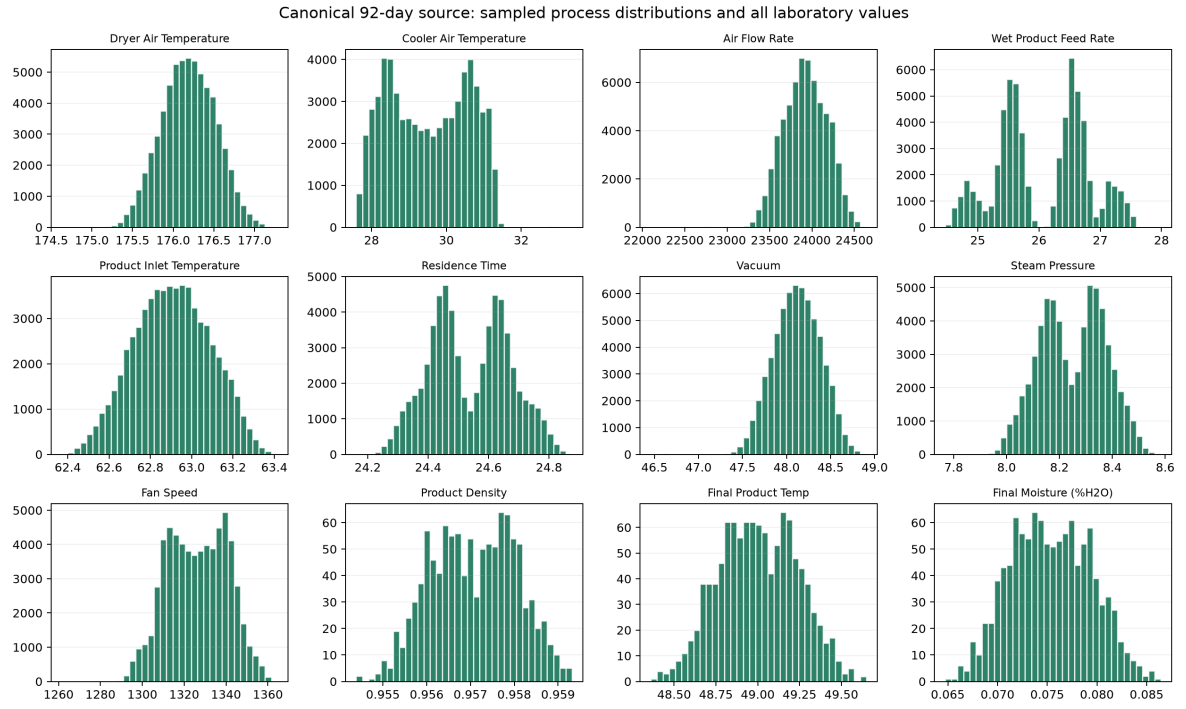
Outlier analysis uses distributions and boxplots to understand the generated operating envelope; unusual points are not deleted automatically because they may carry useful anomaly information. The following plots are generated from the canonical source and its aligned laboratory handoff used by Notebook 03.

Most variables occupy compact, broadly unimodal ranges in this generated dataset, while cooler-air temperature and vacuum show wider or asymmetric shapes. Final moisture is concentrated near the centre of its simulated range with thinner tails, so rare extremes are represented by relatively few rows. These shapes support scale-aware preprocessing and caution against treating the dataset envelope as an approved plant operating range.

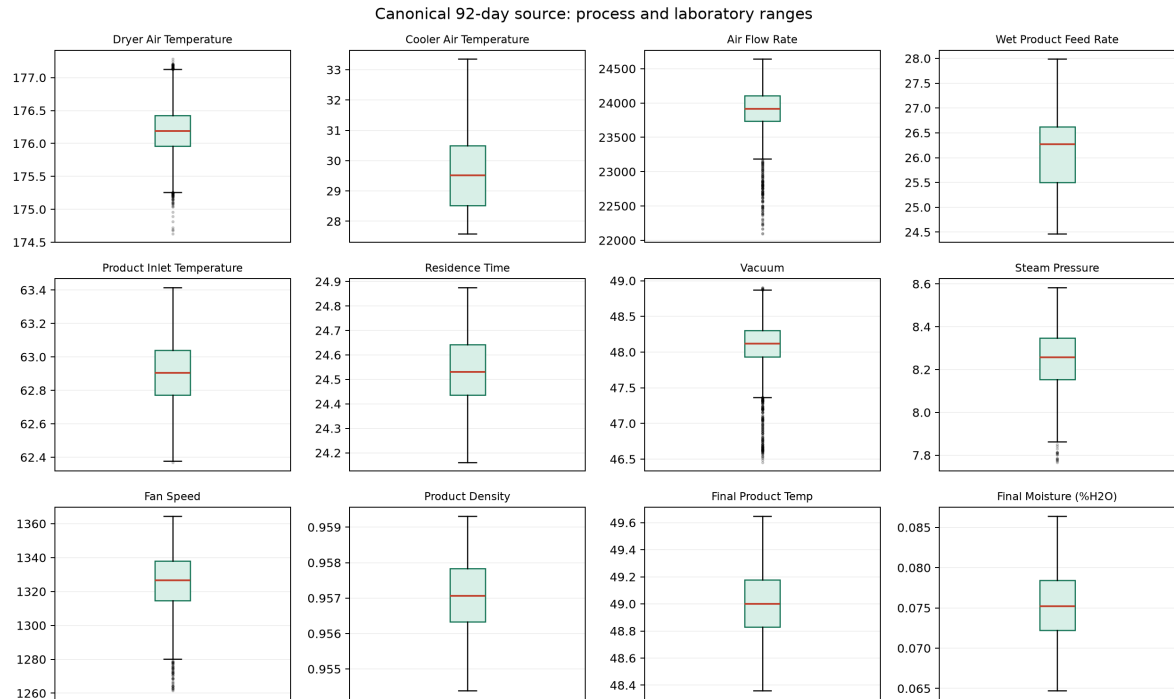
The boxplots show isolated observations for air flow, residence time, vacuum, steam pressure, fan speed and the final-quality fields. They are sparse rather than pervasive, and the plots alone do not distinguish a valid transient from a generated acquisition anomaly. The pipeline therefore retains them for modelling and novelty analysis unless a separate data-quality rule invalidates the observation.

The generator implements causal inertia, gradual drift, multiple regimes and seven labelled ramp–hold–recovery disturbances. Its audited relationships include the expected strong fan-speed/air-flow association ( $r = 0.938$ ), air-flow/vacuum association ( $r = 0.979$ ), inverse feed/residence-time association ( $r = -0.935$ ), inverse dryer-temperature/moisture association at laboratory rows ( $r = -0.508$ ), and cooler-temperature/final-product-temperature association ( $r = 0.909$ ). These are synthetic correlations designed to make the integration exercise physically coherent; they are not

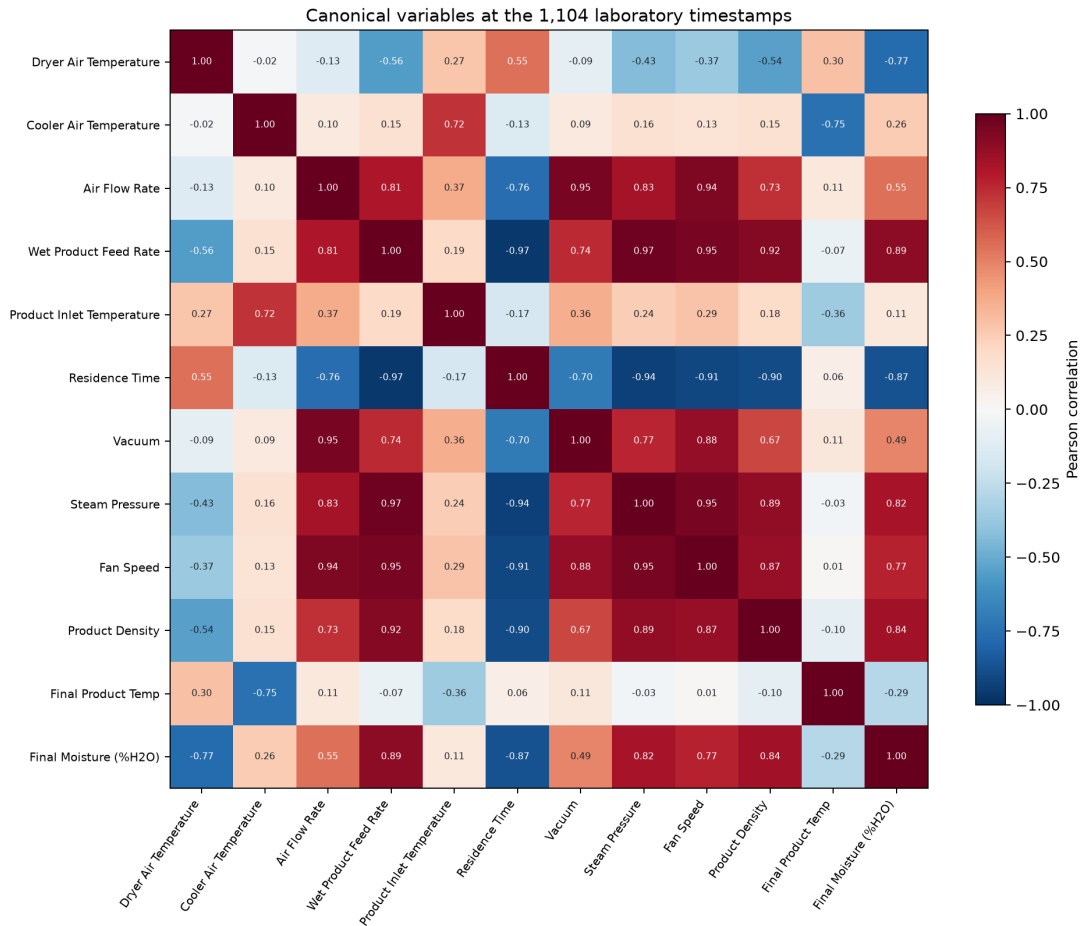




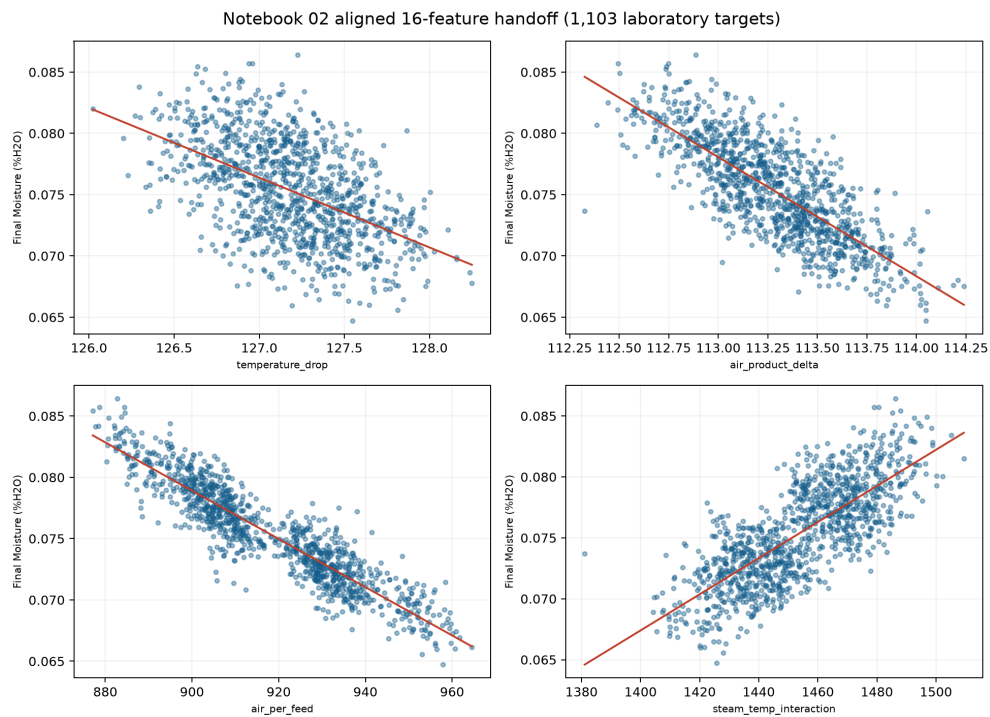
**Figure 4.1:** Distributions of process and sparse quality variables in the canonical source.  
*Source: Executed Notebook 01 output; Author's work.*



**Figure 4.2:** Feature ranges and potential outliers in the canonical source.  
*Source: Executed Notebook 02 output; Author's work.*



**Figure 4.3:** Pearson correlation matrix for canonical process and sparse quality variables.  
*Source: Executed Notebook 02 output; Author's work.*



**Figure 4.4:** Selected aligned engineered-feature relationships with final moisture.  
*Source: Executed Notebook 02 output; Author's work.*

identified plant causal effects or approved operating relationships.

#### 4.2.1 Data-quality risks and treatment decisions

Five categories are separated in the pipeline. A *structural blank* is expected because no lab sample was taken; it remains blank. A *missing process value* prevents the affected instant inference and is reported. A *timestamp gap* is quantified against the chosen five-second prototype grid; it is not hidden by interpolation at ingestion. A *duplicate timestamp* violates the source contract, while a repeated database delivery is handled later by an idempotent upsert. Finally, an *unusual but valid value* is retained for anomaly analysis unless process expertise identifies it as an acquisition error. This classification avoids a single generic “fill missing values” step.

Units and semantics are equally important. The moisture field is stored in percentage points: for example, 0.075 means 0.075% H<sub>2</sub>O, not a fraction equal to 7.5%. Vacuum uses the project’s signed mmH<sub>2</sub>O convention, where more negative means stronger depression. Ratios and interactions are explicitly labelled derived rather than pretending to be directly measured instruments. These decisions are recorded in configuration so that Python, PostgreSQL and Power BI use the same interpretation.

The supplied synthetic file provides a clean cadence, but a future historian extract will require additional checks: tag-quality status, frozen values, shutdown periods, grade/campaign identifiers, manual modes, calibration changes and clock alignment with the laboratory information system. Removing these states indiscriminately would bias the model toward stable operation; retaining them without labels could confuse normal operating modes with faults. The data-preparation review must therefore be conducted jointly with process, laboratory and automation stakeholders.

The variables also differ strongly in numerical scale: fan speed and air flow are orders of magnitude larger than pressure or residence time. Standardization is therefore fitted inside each training fold for the SVM and regularized regression paths. All displayed ranges belong only to the supplied synthetic dataset; none is presented as a plant target, alarm or approved normal boundary.

### 4.3 Residence-time-aware alignment

The most important leakage-control decision is the temporal relationship between a later laboratory sample and the material that passed through the dryer earlier. For a lab result at time  $t_{lab}$ , the direct process snapshot is selected at

$$t_{effective} = t_{lab} - t_{residence}(t_{lab}) - t_{transport}. \quad (4.1)$$

The final configuration uses the most recent process row at or before  $t_{effective}$  and a configurable transport delay set to zero minutes for the generated artifacts [9]. It does not replace the original variables with window means, standard deviations or final-value suffixes. Missing residence time rejects the prediction rather than falling back to a future or arbitrary row.

Binary timestamp searches locate the shifted process row and the strictly previous laboratory input efficiently on the 1.59-million-row source:

**Listing 4.2:** Residence-aware direct-snapshot construction.

```
effective_time = effective_window_end(
    lab_timestamp, residence, transport_delay_minutes
)
process_row = process_sorted.loc[
```

```

    process_sorted["Timestamp"] <= effective_time
].iloc[-1]
previous_lab = lab_sorted.loc[
    lab_sorted["Sample Timestamp"] < lab_timestamp
].iloc[-1]

```

Although the optional exploratory feature engine can calculate 103 candidate window quantities, the final Notebook 02 handoff does not use their mean/std/last columns. It retains the original 16 named inputs: nine direct dryer variables; the most recent strictly previous laboratory density and final-product-temperature values; and `temperature_drop`, `air_product_delta`, `air_per_feed`, `steam_temp_interaction` and `heating_index`. The direct process row is selected at the residence-adjusted time, so no future process observation enters the target row. Of 1,104 laboratory observations, 1,103 form supervised rows; the initial timestamp has no strictly previous laboratory input. The split is chronological: 772 TRAIN, 166 VALIDATION and 165 TEST rows. Five expanding folds tune inside TRAIN, VALIDATION selects the model family, and TEST is opened once. Runtime applies the same ordered feature and prior-laboratory rule [10, 18].

#### 4.3.1 Feature families and their intended meaning

Window statistics remain available for exploratory analysis, but they are not inputs to the final moisture model. Its operating point is represented by the direct process snapshot at the residence-adjusted time, while density and final product temperature use only the latest strictly earlier laboratory result.

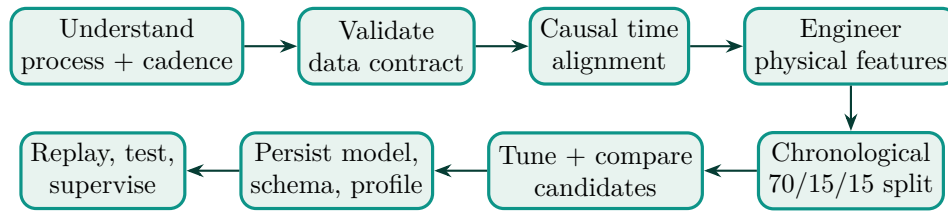
The five retained interactions reduce the burden on a linear model. Temperature drop relates dryer air to the last known final product temperature; air/product delta represents the inlet thermal difference; air per feed relates transfer capacity to wet loading; steam-temperature interaction combines thermal supply and dryer-air level; and heating index combines residence time with dryer-air temperature. Each remains a proxy because the dataset lacks complete mass, composition and energy measurements.

Feature construction occurs after causal time alignment and before model fitting. Scaling parameters are learned on training rows only. Candidate selection sees the validation segment, while the test segment remains untouched until the final model is fixed. This ordering prevents three leakage forms: future process or laboratory rows entering a target, validation/test statistics entering preprocessing, and repeated candidate decisions being made on the final test set.

The 16-feature contract relative to 772 training labels is more controlled than the earlier wide prototype, but effective sample size remains a model-risk concern. Regularization, simple candidates and chronological validation reduce but do not eliminate it. For industrial data, feature stability should be examined across campaigns and rolling-origin evaluation should quantify selection variance. A high process-row count does not remove this concern because supervised information still arrives only at lab times.

#### 4.4 Instant features for anomaly monitoring

At each step of the chosen five-second prototype replay, anomaly inference uses nine process variables plus six instant engineered quantities: air/product temperature difference, cooler difference, air/feed ratio, steam/feed ratio, thermal load and thermal exposure. Protected denominators and required-input checks produce a `DATA_QUALITY` state when needed; standardization and a saved feature schema lock the order and version. This anomaly contract remains process-only. The separate



**Figure 4.5:** Machine-learning workflow embedded in the engineering development cycle.  
*Source: Author's work from the MAP-Dryer-AI repository.*

moisture contract uses the latest strictly previous sparse density and product-temperature result and reports prediction unavailable until such a prior result exists.



# AI Models: Moisture Prediction and Anomaly Diagnosis

## 5.1 Two complementary supervision functions

The project contains two principal modelling components. The supervised soft sensor estimates final moisture, while the unsupervised component examines every row of the chosen prototype replay grid for novelty relative to a reference regime. A deterministic diagnostic layer then translates an anomalous observation into physical evidence. Prediction answers “what moisture is expected?”; novelty detection answers “does the operating pattern resemble the reference?”; diagnosis answers “what should be checked first?”

## 5.2 Chosen moisture model and evidence contract

Notebook 03 is the single source of truth for both the reported moisture experiment and the runtime soft sensor. It reads `data/processed/MAP_Dryer_Lab_Aligned_16.csv`, the 1,103-row handoff produced by Notebook 02 from the canonical 92-day source. The data are split chronologically into 772 TRAIN, 166 VALIDATION and 165 untouched TEST rows. The runtime subsequently computes and orders the same 16 original named features and loads the exact Notebook 03 artifact.

The contract uses nine direct process variables, the five engineered features defined in Notebook 02, and the most recent strictly previous laboratory density and final-product-temperature values. Final moisture remains the target and is never an input. Standardization is fitted inside each training fold. Five-fold expanding `TimeSeriesSplit` on TRAIN tunes candidate hyperparameters, the later VALIDATION segment selects the model family, and TEST is opened once after selection and refitting on TRAIN plus VALIDATION.

**Table 5.1:** Audited evidence contract for the chosen Notebook 03 moisture model

| Item              | Audited value                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Dataset           | MAP_Dryer_Lab_Aligned_16.csv; 1,103 laboratory-aligned rows derived from the canonical source                                 |
| TRAIN             | 772 rows (69.99%); 16 Apr. 2026 02:00–19 June 2026 08:00                                                                      |
| VALIDATION        | 166 rows (15.05%); 19 June 2026 10:00–3 July 2026 04:00                                                                       |
| TEST / replay     | 165 labels (14.96%); 3 July 2026 06:00–16 July 2026 22:00; no shuffle                                                         |
| Selection         | Five-fold expanding <code>TimeSeriesSplit</code> inside TRAIN; model family chosen by VALIDATION RMSE                         |
| Retained pipeline | Standardization → Ridge Regression                                                                                            |
| Hyperparameters   | Ridge $\alpha = 10$                                                                                                           |
| Model size        | Exactly 16 original ordered input features                                                                                    |
| Artifact          | <code>models/5s/quality_moisture_pipeline.joblib</code> ; metadata in <code>artifacts/notebook03_model_evaluation.json</code> |

*Source: Executed Notebook 03 and its saved evaluation record [11].*

The chronological split is explicit and reproducible:

**Listing 5.1:** Notebook 03 chronological evidence boundaries.

```

n_train = round(len(aligned) * 0.70)
n_validation = round(len(aligned) * 0.85) - n_train
train = aligned.iloc[:n_train]
validation = aligned.iloc[n_train:n_train + n_validation]
test = aligned.iloc[n_train + n_validation:]
tscv = TimeSeriesSplit(n_splits=5) # TRAIN only

```

### 5.3 Candidate comparison and Ridge selection

Five candidates are evaluated under the same forward-looking structure. Ordinary Linear Regression provides a transparent reference. Ridge tests L2 shrinkage. Elastic Net tests combined L1/L2 regularization, while Random Forest and Gradient Boosting test nonlinear mappings. Each candidate's tuning remains inside TRAIN; its fitted best configuration is then scored once on VALIDATION.

Ridge gives the lowest VALIDATION RMSE, 0.0013566 percentage points, narrowly ahead of Elastic Net at 0.0013625 and Linear Regression at 0.0013671. Selection is therefore supported by a later unseen segment, not by the final TEST score. Notebook 03 also contains separate Elastic Net training and metrics cells, matching the other baseline models. The selected tuning structure is:

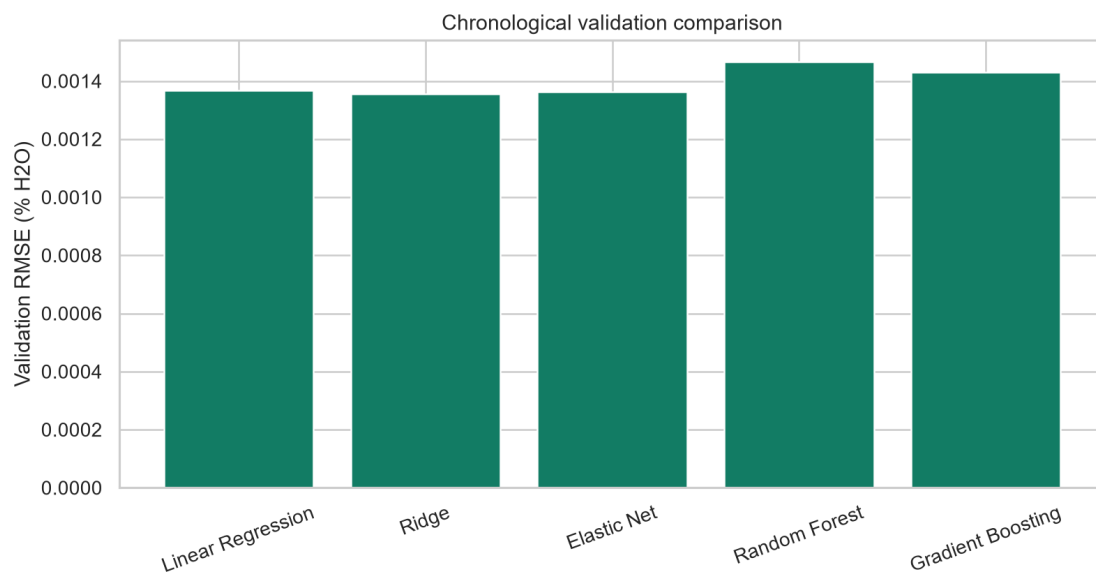
**Listing 5.2:** Leakage-safe Ridge pipeline and tuning grid.

```

ridge = Pipeline([
    ("scaler", StandardScaler()),
    ("model", Ridge()),
])

search = GridSearchCV(
    ridge,
    param_grid={"model__alpha": [0.01, 0.1, 1, 10, 100]},
    cv=TimeSeriesSplit(n_splits=5),
    scoring="neg_root_mean_squared_error",
)

```



**Figure 5.1:** Later-segment VALIDATION RMSE used to select Ridge after training-only tuning.  
*Source: Executed Notebook 03 figure output [11].*

The Ridge/Elastic Net margin is only about 0.43%, so this is a versioned choice for the canonical dataset and declared split, not a universal claim of model-family superiority. The selected Ridge

mean training-fold CV RMSE is 0.0014050; that diagnostic tunes its hyperparameters, while the separate later VALIDATION score governs family selection.

## 5.4 Untouched holdout performance

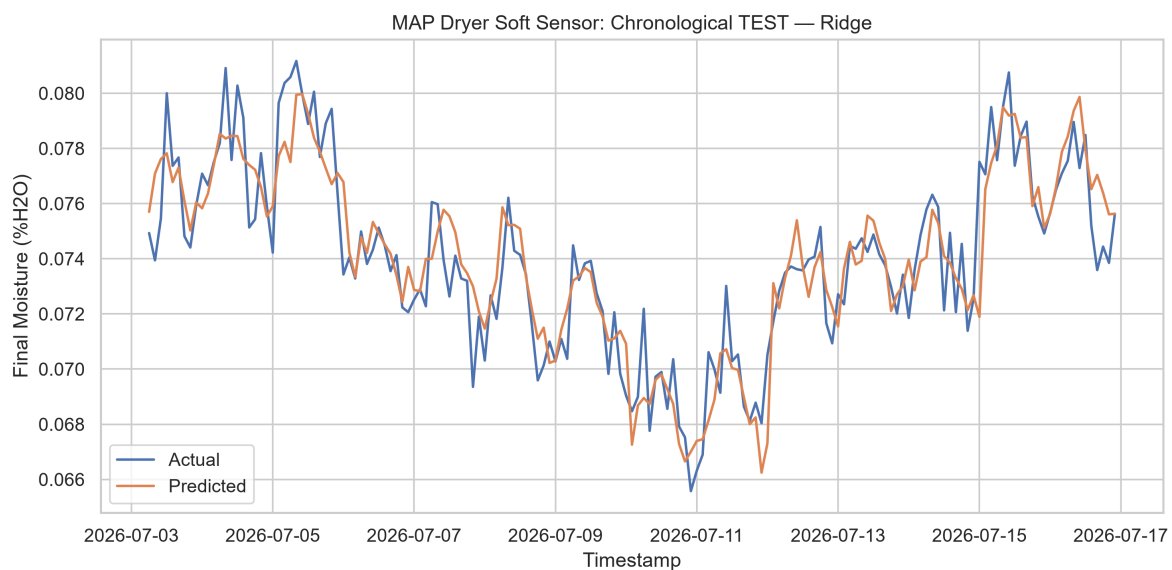
After the model family and hyperparameters are frozen, the pipeline is refitted on the combined 938 TRAIN and VALIDATION rows and evaluated once on the 165-row TEST tail. It achieves MAE 0.001069, RMSE 0.001403,  $R^2 = 0.824530$ , bias 0.000066 and maximum absolute error 0.005621 percentage points  $H_2O$ .

**Table 5.2:** Selected Ridge performance across the three evidence partitions

| Partition      | Samples | MAE             | RMSE            | $R^2$           |
|----------------|---------|-----------------|-----------------|-----------------|
| TRAIN          | 772     | 0.001087        | 0.001414        | 0.878173        |
| VALIDATION     | 166     | 0.001031        | 0.001357        | 0.895937        |
| Untouched TEST | 165     | <b>0.001069</b> | <b>0.001403</b> | <b>0.824530</b> |

Source: Executed Notebook 03 [11]; errors are moisture percentage points.

Error magnitude is stable across the three chronological segments: TEST RMSE is about 0.8% below TRAIN RMSE. TEST  $R^2$  is lower than TRAIN by about 0.0536, but the untouched score remains approximately 0.825; the supplied synthetic target still retains variation that the 16-feature regularized linear contract does not explain.



**Figure 5.2:** Actual and predicted moisture over the untouched chronological TEST interval.

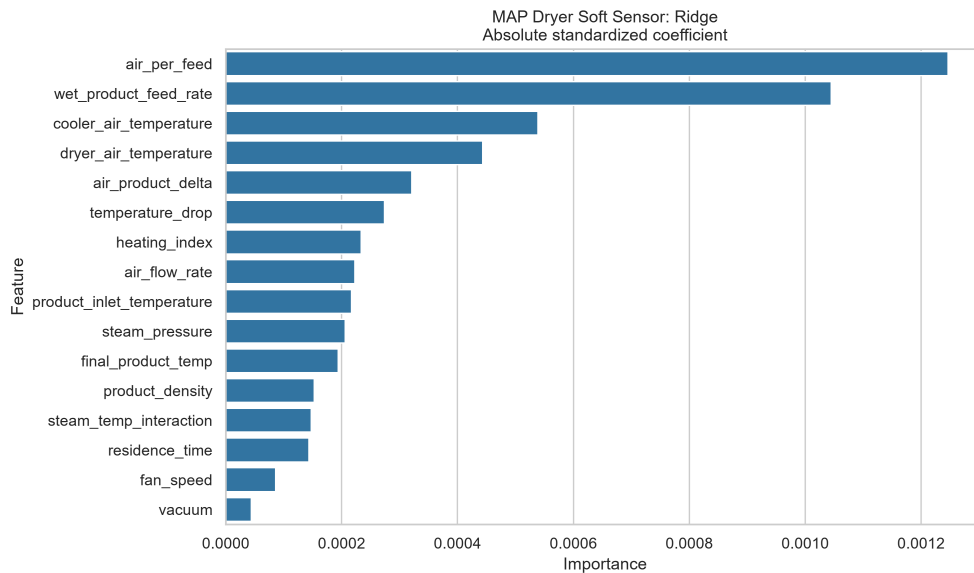
Source: Executed Notebook 03 figure outputs [11].

Predictions follow the moisture variation but remain smoother than the laboratory series; the small positive TEST bias indicates slight overprediction on average. The maximum error remains larger than the mean error, reinforcing that this synthetic result is a software-integration benchmark and an advisory soft sensor requiring laboratory confirmation.

## 5.5 Model interpretation and risk

The Ridge model consumes all 16 governed inputs. Figure 5.3 reports absolute standardized coefficient magnitudes.

Air per feed and wet-product feed rate have the largest absolute standardized coefficients, followed



**Figure 5.3:** Absolute standardized coefficients of the selected Ridge pipeline.

*Source: Executed Notebook 03 figure outputs [11].*

by cooler-air and dryer-air temperature. Correlated inputs can redistribute weight, and coefficient magnitude does not establish causation.

$R^2$  describes explained variation in this particular holdout sequence; it does not show whether errors are acceptable against an approved product specification. MAE describes average absolute deviation, RMSE penalizes larger misses, and the time/residual plots expose local behaviour. Industrial validation must also quantify analytical uncertainty, performance by campaign and grade, uncertainty coverage, drift, and errors near approved product limits.

L2 shrinkage controls the correlated 16-feature contract, yet it cannot compensate for an unrepresentative dataset or extrapolation outside training support. Any real-data study must rebuild the comparison rather than assume that Ridge or this hyperparameter remains optimal.

### 5.5.1 Notebook model and dashboard runtime consistency

Notebook 03 saves `models/5s/quality_moisture_pipeline.joblib`, and the runtime loads that exact file. The artifact records the same ordered 16 feature names exposed by the schema; startup validation rejects any mismatch. The replay source is the 237,600-row TEST interval derived from the canonical source, so PostgreSQL and Power BI receive inference made by the same model evaluated above rather than a separately trained dashboard model.

## 5.6 Novelty detection and risk display

Notebook 04 is the authoritative source for the anomaly runtime export. Its final cells use a One-Class SVM with an RBF kernel, `gamma='scale'` and  $\nu = 0.02$  [12, 19]. It is fitted on a deterministic 6,000-row sample from the 1,111,681 process rows in the canonical TRAIN period and checked on a separate 20,000-row validation sample. The observed validation flag rate is 5.925%. This is a novelty-rate diagnostic on supplied synthetic process patterns, not a false-positive rate or fault-detection accuracy because no confirmed plant fault labels exist [10].

The raw decision function is not a probability. A logistic mapping converts it to a display risk in  $[0, 1]$ , with 0.5 at the detector boundary and 0.8 used as a high-risk display threshold. This improves dashboard readability but does not create a statistically calibrated failure probability.

**Listing 5.3:** Final service mapping from SVM output to display risk.

```

score = float(anomaly_model.decision_function(scaled)[0])
anomaly_detected = int(anomaly_model.predict(scaled)[0]) == -1
risk = 1.0 / (1.0 + math.exp(
    min(max(score / risk_scale, -60.0), 60.0)
))

```

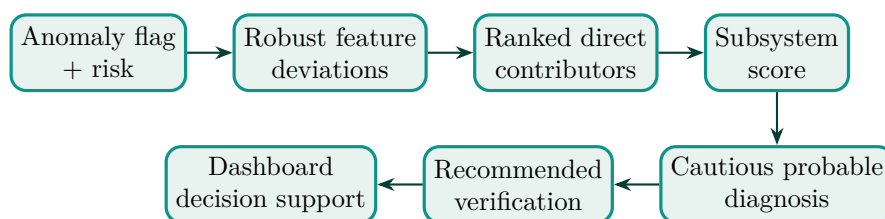
For a normalized feature vector  $\mathbf{x}$ , the RBF One-Class SVM learns a boundary around the reference observations. The scikit-learn decision function is positive on the accepted side and negative on the novelty side; the service consequently raises the anomaly flag below zero. The parameter  $\nu = 0.02$  acts as a design control on training errors/support vectors rather than a guaranteed future anomaly rate [19]. The observed validation flag rate is therefore reported, not assumed.

The reference profile and SVM serve different purposes. The SVM evaluates the joint multivariate pattern. Robust univariate quantiles identify which individual features are unusual enough to explain the event. A row can be novel even when no single raw variable exceeds a percentile limit, because the combination is unusual; in that situation the service assigns low severity and states that no direct abnormal contributor was found. Conversely, a raw variable outside its 99th percentile supports a high operational-severity display only when the multivariate detector has also flagged the observation.

## 5.7 From anomaly to operator-readable diagnosis

The real-time diagnostic engine uses a train-period reference profile containing median, MAD, IQR and 1st/5th/95th/99th percentiles per feature. A robust scale is computed as the maximum of scaled MAD, scaled IQR and a numerical floor. Each feature receives a direction and variable severity: values inside the 5th–95th percentile band are normal, values between the 1st/5th or 95th/99th limits are medium deviations, and values beyond the 1st or 99th percentile are high deviations. Robust distance is multiplied by feature criticality to rank evidence.

Engineered features participate in subsystem voting, while only raw process features are returned as operator-facing contributors. A feature associated with multiple subsystems divides its score among them. The selected subsystem is the one with the greatest accumulated contribution. A transparent rule map produces a cautious diagnosis and recommended verification for the leading direct contributor. Event severity is **LOW** when no direct abnormal contributor exists, **MEDIUM** for at least two medium deviations, and **HIGH** when at least one direct feature exceeds its 1st/99th percentile range. **CRITICAL** is deliberately not assigned because statistical quantiles are not approved alarm or trip limits [13, 7].



**Figure 5.4:** Anomaly-to-diagnosis transformation implemented for real-time supervision.  
*Source: Author's work from the MAP-Dryer-AI repository.*

The broader configuration-driven diagnostic work defines eight concise rule families: insufficient drying air, insufficient thermal supply, dryer overload, reduced cooling, abnormal exhaust/pressure, possible temperature-sensor bias or jump, possible stuck sensor, and process instability or oscillation

[13]. Air/fan evidence leads to supply and measurement checks; steam/thermal evidence leads to heating checks; feed, residence and ratio evidence leads to loading checks; and cooler, vacuum or temporal evidence directs the remaining subsystem checks. These rules informed validation scenarios, but the final replay service uses its dedicated quantile/ranking engine. This distinction prevents evidence from one implementation path being attributed to another.

These rules are deliberately phrased as hypotheses. A low air-flow reading can represent a true delivery problem, a closed damper, an exhaust interaction or a biased instrument. The display therefore separates probable diagnosis, possible causes and recommended verification. In a future feedback loop, confirmed causes could be used to evaluate precision by rule; the current synthetic scenarios test logical acceptance, not empirical causal accuracy.

## 5.8 Artifacts, traceability and failure behaviour

One governed artifact set is deliberately visible. Notebook 03 saves the selected moisture pipeline and JSON evaluation record; Notebook 04 saves the anomaly model, scaler, reference profile and its evaluation record. The runtime loads those exact governed artifacts from `models/5s`, while the shared schema records both ordered feature contracts. Runtime versions travel with database outputs and feature order is checked at inference. Invalid or incomplete input produces `DATA_QUALITY`; insufficient shifted history produces an unavailable prediction with a reason. These are essential production-engineering behaviours: a system must reveal which compatible artifact it used and when it could not reason safely.

### Meaning of explainability

The diagnostic output is evidence localization, not proven causality. “Likely subsystem” means that abnormal, criticality-weighted features assigned to that subsystem dominate the current observation. “Recommended verification” tells the operator what to inspect; it is not a command or maintenance conclusion.



# Real-Time Prototype Architecture: Python, PostgreSQL and Simulation

## 6.1 Repository structure and technical responsibilities

The implementation is organized so that experimentation, runtime logic and presentation remain traceable. Notebooks 01–04 form an executable dependency chain: canonical audit, residence-aware 16-feature handoff, chronological moisture-model selection/export, then anomaly/diagnosis export. Reusable modules under `src/multirate` implement preprocessing, instant features, temporal alignment and the wider window-feature engine, while `src/canonical_pipeline.py` governs the shared data and model contracts. `src/diagnosis` contains the configurable diagnostic work. The `realtime_pipeline` directory contains database bootstrap scripts, the held-out replay/inference service and verification utilities. The PBIP project and its TMDL semantic model are stored under `POWERBIDASHBOARD` [7].

This separation supports traceability without duplicating the models. Notebook 03 exports the moisture artifact and Notebook 04 exports the anomaly artifacts that the service loads. SQL views stabilize the contract presented to Power BI. Tests exercise the modules without requiring the dashboard, while the live backend validator compares SQL view columns with the physical `sourceColumn` bindings in TMDL.

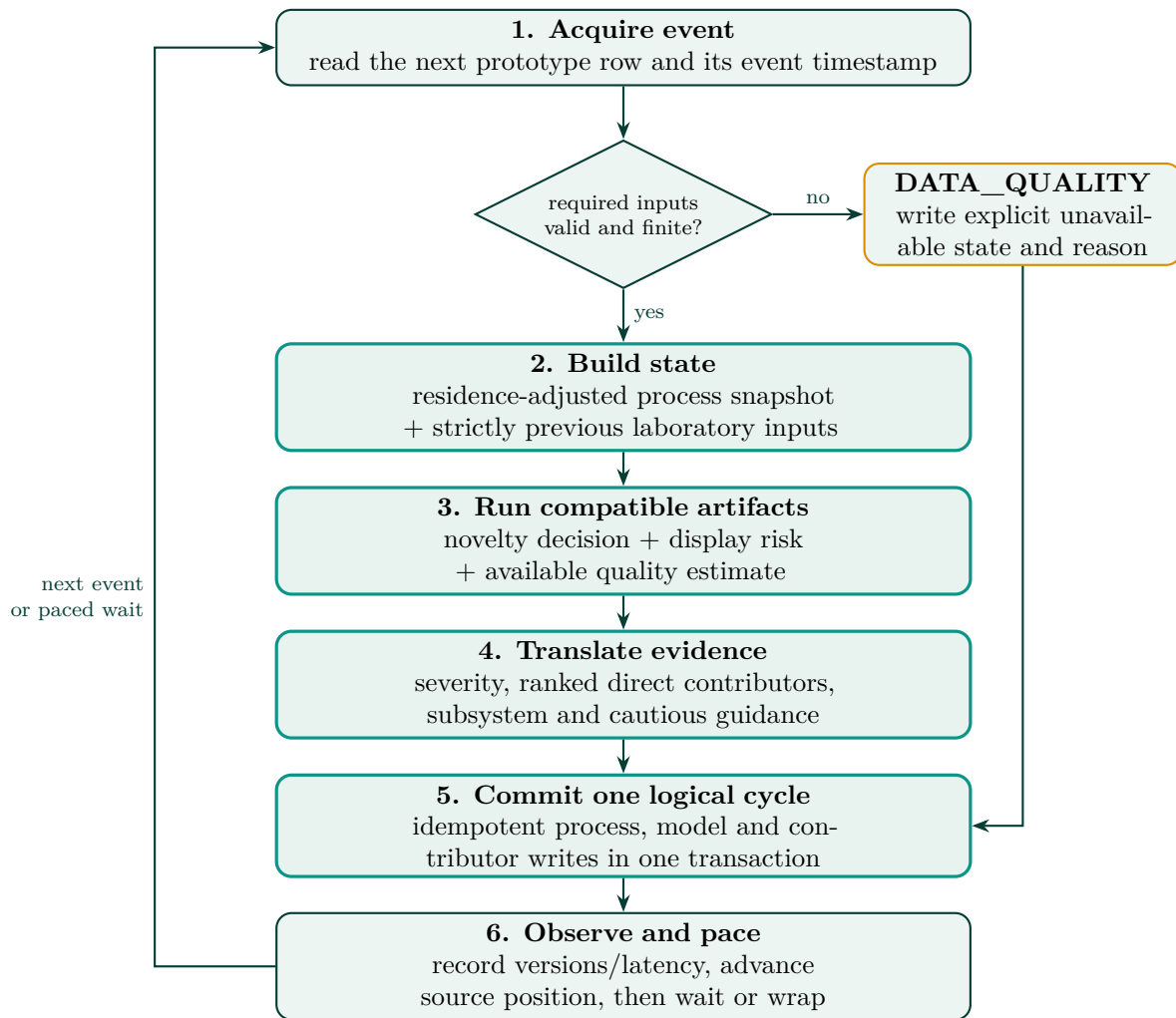
## 6.2 Prototype replay and inference loop

In the plant context, process variables are available continuously through the monitoring and control environment. The service does not reproduce the native acquisition layer: it replays the canonical held-out TEST interval containing 237,600 rows from 3–16 July 2026. Its default pacing represents one event every five seconds, while bounded verification can disable waiting without changing event timestamps. At each cycle it validates the row, selects the direct process snapshot at the residence-adjusted time, carries only the most recent strictly previous density/product-temperature result into the 16-feature moisture contract, runs the Notebook 03 moisture model and Notebook 04 anomaly artifacts, and stores the outputs. Laboratory fields remain blank except at 165 configured sparse sample timestamps. Model, feature schema and source are tied together by version and SHA-256 metadata.

The executable loop keeps inference, persistence and pacing separate:

**Listing 6.1:** Core replay, inference and pacing sequence.

```
cycle_started = time.monotonic()
source_row = source.iloc[next_position].copy()
new_row = source_row[["Timestamp", *PROCESS_VARIABLES]].to_frame().T
buffer = pd.concat([buffer, new_row], ignore_index=True).tail(buffer_rows)
inference = run_row_inference(source_row, buffer, artifacts, config)
latency_ms = (time.monotonic() - cycle_started) * 1000.0
connection = persist_with_retry(
    connection, source_row, inference, artifacts, latency_ms
)
```



**Figure 6.1:** Event-level workflow paced at the selected five-second prototype interval.

*Source: Author's work from the MAP-Dryer-AI repository.*

```

elapsed = time.monotonic() - cycle_started
target = config.poll_interval_seconds / config.replay_speed
time.sleep(max(0.0, target - elapsed))

```

The buffer is long enough to locate the residence-adjusted process row and retain the most recent strictly previous laboratory density and product-temperature result. Missing residence time, a missing shifted process row or unavailable prior laboratory inputs produces an unavailable moisture estimate rather than an improvised value. Database upserts make a retried timestamp safe, and each cycle records latency against the selected five-second demonstration budget.

### 6.3 PostgreSQL persistence layer

The database is a traceability and integration boundary. `dryer_map` stores the date/time key, nine process fields and three nullable quality fields. `dryer_model_outputs` stores prediction, anomaly score/risk, status, severity, subsystem, diagnosis, causes, verification, model/schema versions and inference latency. `dryer_abnormal_variables` stores ranked contributor evidence. Stored functions wrap quoted legacy identifiers and implement `ONCONFLICT` upserts keyed by date and time [14].

The main dashboard view uses a lateral as-of lookup to expose the latest preceding lab result and

**Table 6.1:** Principal database objects and dashboard roles

| Object                                        | Purpose                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------|
| <code>dryer_map</code>                        | Process and sparse laboratory observations; composite timestamp key prevents duplicates. |
| <code>dryer_model_outputs</code>              | Versioned prediction, novelty, diagnosis and latency output per processed timestamp.     |
| <code>dryer_abnormal_variables</code>         | Direct contributor rank, deviation, reference limits and variable severity.              |
| <code>vw_dryer_dashboard_powerbi</code>       | Joined current-grain facts plus latest-lab as-of context and validated error.            |
| <code>vw_dryer_lab_samples</code>             | Only true lab rows, with prediction-at-sample and error.                                 |
| <code>vw_dryer_overview_trends_powerbi</code> | Lightweight latest eight-hour trend for frequent dashboard queries.                      |
| <code>vw_dryer_anomaly_events</code>          | Groups anomalous rows separated by no more than 30 seconds into operator events.         |
| <code>vw_dryer_contributors_powerbi</code>    | Dashboard-facing contributor evidence.                                                   |

*Source: Database bootstrap and prototype-grid upgrade scripts [14].*

its age. This is semantically useful but relatively expensive over long trends; the separate overview view therefore restricts itself to the latest eight hours and avoids repeating the as-of query for every chart point. Timestamp and model-version indexes support the frequent filters. This is a performance decision derived from dashboard workload, not a generic database embellishment.

**Listing 6.2:** Simplified event grouping used by the anomaly-events view.

```
CASE WHEN ts - LAG(ts) OVER (ORDER BY ts) <= INTERVAL '30 seconds'
      THEN 0 ELSE 1 END AS is_new_event
-- cumulative sum of is_new_event gives one event identifier
```

The service writes one logical cycle inside a transaction. If model inference succeeds but a later database operation fails, the transaction can be rolled back rather than leaving incomplete process/model pairs. Credentials are read from an environment file and are not reproduced in this report.

## 6.4 Power BI-facing semantic layer

Five DirectQuery tables are generated in the semantic model: dashboard, contributors, lab samples, anomaly events and overview trends. Relationships connect contributor/event context and laboratory context while measures provide current values, state, age and formatted labels. The connection remains query-based, so the active Power BI page requests current rows from PostgreSQL at each refresh rather than importing a scheduled snapshot.

Microsoft documents automatic page refresh as a DirectQuery feature and notes that effective cadence depends on query duration, data-source load and service capacity settings [20]. The PBIP configures a five-second fixed interval in Desktop as the visualization target for this prototype. It is neither a claim about plant measurement frequency nor a guarantee for every Power BI Service workspace; deployment requires gateway, capacity and load qualification.

## 6.5 Prototype execution evidence

The final backend smoke test processed the first 1,500 rows of the held-out TEST replay and its two sparse laboratory observations. It loaded model version `canonical_92d_v6.0`, produced no anomaly flags in this initial interval, and measured 9/47 ms average/maximum cycle latency.

The service then stopped at the configured `MAX_CYCLES` bound. This deliberately bounded run establishes model loading, causal prior-laboratory state, transactional writes and live Power BI view compatibility; it is not a claim that the entire 237,600-row TEST replay was executed against PostgreSQL or that deployment performance is qualified [15].

These numbers measure Python/database processing under prototype conditions. They exclude a live PCS7 connector, plant network delays, concurrent Power BI user load, service-capacity throttling and production resilience. The correct conclusion is that the implementation has substantial timing margin for controlled replay, not that a future industrial system is already performance-qualified.

## 6.6 Failure handling, observability and deployment concerns

The service rejects missing or non-finite inputs and records `DATA_QUALITY` instead of forcing a model result. An unavailable shifted snapshot or prior laboratory input carries a reason, transactions keep related writes consistent, and cycle latency, source position and model state are logged. This separates “the process looks abnormal” from “the analytics could not evaluate the process.”

The hardened prototype uses environment-provided credentials, structured connection parameters, explicit SSL mode, finite replay and documented example roles for a runtime writer and read-only Power BI consumer. It retains naïve plant-local timestamps only for compatibility. Industrialization must still add managed secrets, network/security approval, UTC ingestion and governed display conversion, buffering/retry limits, recoverable PostgreSQL operation and model rollback. Qualification should measure source age, p95/p99 inference/write/query latency and concurrency through the gateway, then test disconnects, reordered or duplicate events and delayed laboratory results. The five-second visualization target is retained only if the complete deployed chain remains stable and makes interruption visible.

### 6.6.1 End-to-end traceability checks

An accepted cycle should be traceable from source to display. The source timestamp and values are first validated against the CSV/process contract. The feature builder records the schema version and constructs the ordered vector. Model outputs carry model version, inference timestamp and latency. PostgreSQL stores the process and model rows under the same date/time key and the contributor rows under the event timestamp. Views then expose a stable set of names and grains to TMDL; Power BI measures locate the latest eligible timestamp and display freshness.

This chain supports practical audit questions. If a card appears stale, one can compare latest process timestamp with inference and query time. If a diagnosis looks implausible, one can retrieve the direct contributors and reference limits at that event. If a prediction changes after retraining, the model/schema version identifies the generating artifacts. If a lab error card changes, the lab-sample view reveals the exact sample and prediction pair. Traceability therefore reduces both debugging time and the risk of silently comparing incompatible states.

Final repository validation covers service logic, bounded held-out replay, idempotent SQL/views and live Power BI physical bindings. Production validation must add full-volume source reconciliation, restarts, duplicate or reordered messages, delayed laboratory arrival, failover, gateway interruption and report concurrency, each with a defined stale/data-quality response. Operational logs retain event time, state transition, model version, latency and error category while excluding credentials, connection strings and personal account names. The chain is executable; the remaining industrialization work lies at the interfaces, governance and validation layers.

# Power BI Supervision Dashboard

## 7.1 Operator-centred dashboard design

The dashboard is the point at which process, quality and model evidence become a supervision aid. Its design priorities are state visibility, hierarchy and restrained alarm colour. Normal status uses green, low/medium attention uses amber, high anomaly uses red and unavailable/data-quality conditions use grey. Colours never replace text: each state is also labelled. The two pages separate routine monitoring from diagnostic investigation so that explanations do not overload the main view.

The PBIP uses a DirectQuery semantic model and a deliberately configured five-second automatic page refresh on both pages. Repository validation reports 86 total visuals, 114 field references and 276 model fields across five tables; all field references, navigation actions, refresh settings, ordering and query budgets resolve [17]. The interval is a prototype visualization target, not the native plant measurement frequency. Deployment to Power BI Service would still require an appropriate capacity/gateway design, credentials and performance measurements under expected concurrency [20].

The two figures in this chapter are the final validated 16:9 project renders generated from the SQL-backed preview workflow. They reproduce the final two-page PBIP information architecture without the Power BI Desktop ribbon, authoring panes or credentials. They are faithful layout/value proxies rather than evidence of a deployed Power BI Service application.

## 7.2 Page 1: Operations Overview

Figure 7.1 presents the overview page. Its PostgreSQL demonstration data come from the held-out chronological TEST interval of the canonical source, not from a separately generated dashboard dataset. The upper cards answer immediate operating questions: current predicted moisture, latest laboratory moisture and age, anomaly risk/status and operational severity. Process tiles expose key thermal, flow, feed, steam and vacuum measurements. Trend panels maintain time context rather than reducing the process to one current value. Laboratory markers remain sparse, while the prediction continues at each prototype replay step.

The captured state deliberately demonstrates freshness handling: the header reports data freshness separately from process status, and the last stored diagnosis remains visible as context. This is not an application failure. It shows that old data are not presented as proof of current operation. The static PROTOTYPE · REPLAY indicator and **Replay / Refresh: 5 s** wording make clear that five seconds is the demonstration cadence, not native plant acquisition.

The latest-lab presentation is deliberately paired with age. Without age, a carried-forward result could appear current. In this synthetic prototype, validated error is computed only where a represented sparse laboratory observation and a corresponding prediction coexist. The principal distinction is therefore visible in both model and interface:

- **predicted moisture:** continuous model estimate;
- **latest laboratory moisture:** last available measured value with timestamp/age;
- **validated error:** prediction minus laboratory value only at a sample timestamp.

DirectQuery measures aggregate only the intended grain. A lightweight eight-hour PostgreSQL view feeds the overview trends, avoiding an expensive latest-lab as-of lookup for every historical point. The report validator also enforces a query-visual budget, which reduces the risk that the selected rapid page refresh launches an unnecessarily large set of source queries.

### 7.3 Page 2: Diagnostics and Root Cause

The diagnostic page in fig. 7.2 supports investigation after an anomaly is raised. A timeline provides event context; filter controls support event selection; status fields show the probable subsystem, diagnosis, causes and verification guidance; contributor tiles display direct abnormal variables with direction and severity. The wording remains explicitly advisory.

The selected synthetic event is labelled medium severity and assigned to the feed-loading subsystem. Wet-product feed rate and product-inlet temperature lead the contributor ranking, while residence time is the only displayed high univariate deviation. This combination demonstrates why the page shows several pieces of evidence and a verification prompt: the subsystem label is a reasoned prioritization, not proof of a root cause.

Consecutive anomalous prototype rows separated by at most 30 seconds are grouped into one event. This limits alarm flooding and gives the operator an episode rather than a list of near-identical timestamps. Severity, subsystem and diagnosis for an event are taken from its peak-risk row. The raw score is retained for technical traceability; the risk scale is used for visual interpretation.

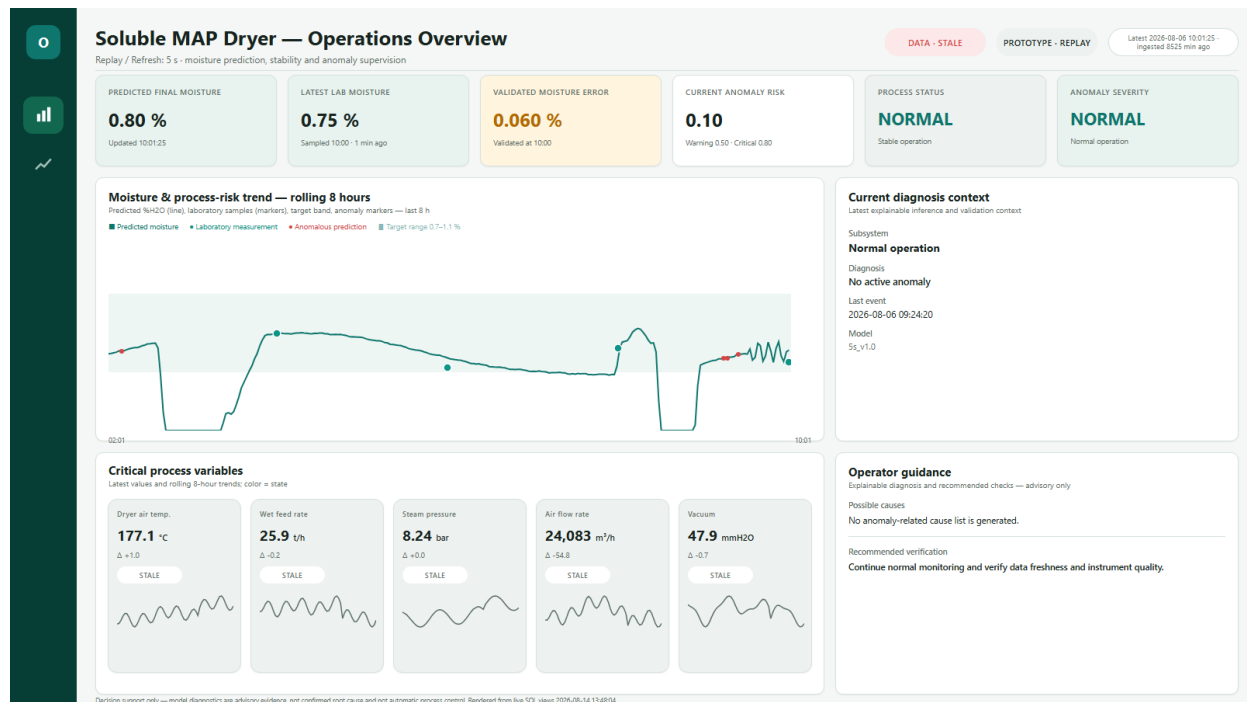
### 7.4 Dynamic status and selected measures

The semantic model groups measures by current values, lab context, validation, risk/status, trends, critical variables and events. The interface answers three questions in order: **what is happening**, through process/moisture/lab state; **where evidence is concentrated**, through direct contributors and subsystem voting; and **what should be verified**, through cautious diagnosis and an inspection prompt. Unavailable information is displayed as a dash or explicit state rather than zero, and DATA\_QUALITY is never styled as an equipment anomaly.

#### 7.4.1 Semantic model and DAX logic

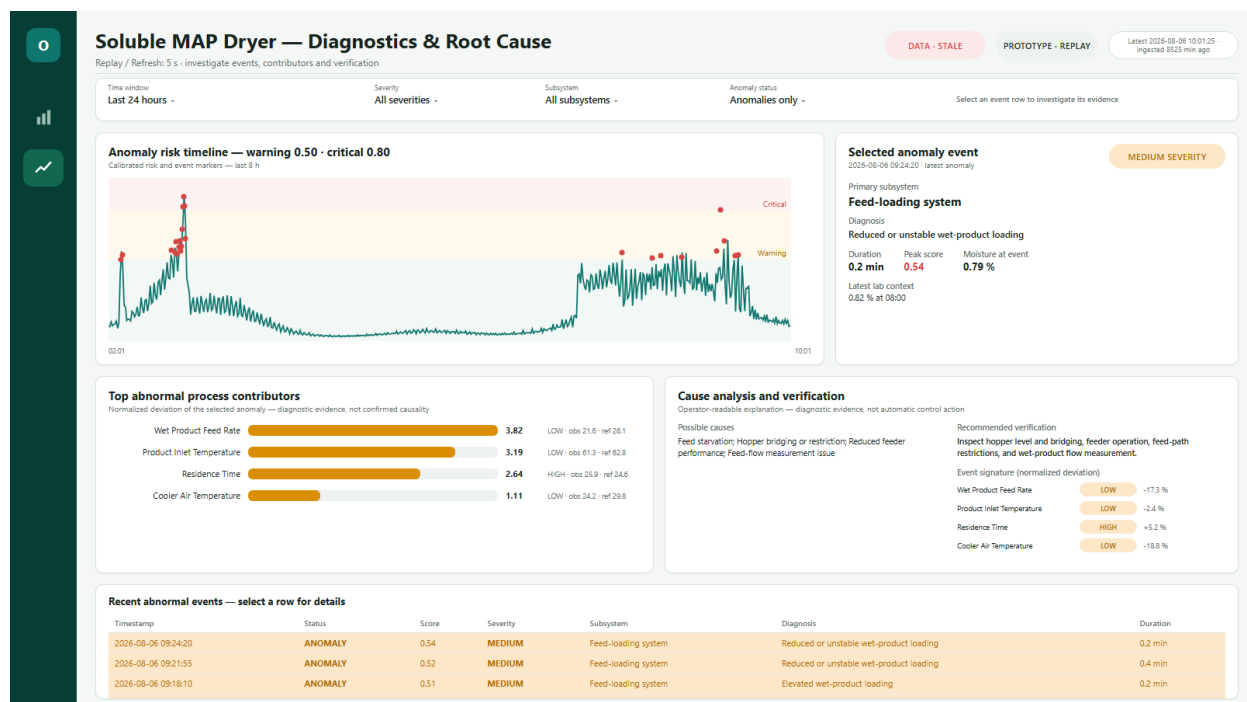
The five semantic tables preserve different grains: one row per process timestamp, recent trend point, true laboratory sample, grouped anomaly event, or ranked event contributor. This prevents contributor joins from multiplying process rows and prevents carried-forward lab context from increasing the sample count. Latest-lab age uses the true lab timestamp and latest process timestamp:





**Figure 7.1:** Final SQL-backed Operations Overview with separate data and process status, moisture supervision and process context.

Source: MAP-Dryer-AI final academic-showcase dashboard render; Author's work.



**Figure 7.2:** Final SQL-backed Diagnostics and Root Cause page with event context, ranked contributors and verification guidance.

Source: MAP-Dryer-AI final academic-showcase dashboard render; Author's work.

**Listing 7.1:** Simplified DAX pattern for laboratory-result age.

```

Lab Result Age Minutes =
IF (
    ISBLANK ( [Lab Last Sample Timestamp] ) ||
    ISBLANK ( [Latest Timestamp] ),
    BLANK (),
    DATEDIFF ( [Lab Last Sample Timestamp],
               [Latest Timestamp], SECOND ) / 60.0
)

```

Freshness has precedence: absent or old data produce **STALE** even if the last stored anomaly state was normal; otherwise anomaly, watch and normal states are evaluated. The palette never replaces the text label. Although a critical colour exists for display compatibility, the Python engine does not assign **CRITICAL**; its reference-profile quantiles are statistical and not approved safety limits.

## 7.5 Illustrative operator-use scenario

In a synthetic replay episode, the operator first checks lab-result age and treats the soft-sensor value as an estimate. The trend is then reviewed for coincident changes in air, feed or steam. After selecting the event, ranked contributors and subsystem voting prioritize a verification. For example, low air flow and fan speed may prompt inspection of fan operation, flow instrumentation and exhaust condition. PCS7, approved limits and field information remain authoritative; the model neither changes a setpoint nor acknowledges an alarm.

**Table 7.1:** Operator interpretation guardrails

| Displayed item     | Appropriate interpretation                       | Inappropriate interpretation               |
|--------------------|--------------------------------------------------|--------------------------------------------|
| Predicted moisture | Soft-sensor estimate requiring lab confirmation. | A new laboratory measurement.              |
| Anomaly risk       | Relative distance from the learned reference.    | Probability of equipment failure.          |
| Likely subsystem   | Area with concentrated abnormal evidence.        | Proven mechanical root cause.              |
| Recommended check  | Prioritized verification prompt.                 | Automatic command or approved instruction. |

## 7.6 Usability and performance considerations

The current pages are suitable for a controlled demonstration. Industrial use requires operator testing, accessible colour/text cues, terminology aligned with the site alarm philosophy, approved limit ownership and review of nuisance events. Model monitoring should track version, lab error, missing inputs, stale duration, event rate and drift; retraining must follow a versioned approval process. PostgreSQL, gateway and report-query p95/p99 latency must be tested under expected concurrency. The five-second visualization refresh is retained only if end-to-end measurements justify it [20].

### Dashboard contribution

The dashboard centralizes the operational narrative without obscuring provenance: process values remain process values, laboratory results retain age, predictions remain estimates, anomaly risk remains statistical and diagnosis remains a verification aid. That semantic discipline is the principal usability achievement of the Power BI layer.

## CHAPTER 8

# Results, Validation, Limitations and Future Work

## 8.1 Consolidated prototype results

The project delivers an end-to-end digital supervision prototype rather than disconnected notebook and runtime experiments. Notebooks 01–04 consume one canonical source in sequence: Notebook 03 exports the selected moisture artifact and Notebook 04 exports the anomaly artifacts. The runtime loads those exact artifacts while replaying the held-out TEST interval. SQL views expose stable dashboard semantics and the PBIP presents an operator workflow. The main quantitative results are summarized in table 8.1.

**Table 8.1:** Consolidated evidence from the final repository

| Evidence                    | Result                                                                           | Correct interpretation                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Selected moisture TEST      | MAE 0.001069; RMSE 0.001403; $R^2$ 0.824530; $n = 165$                           | Ridge with the original 16 named features; supplied synthetic prototype evidence only.                                      |
| Moisture model selection    | VALIDATION RMSE 0.0013566                                                        | Five expanding folds tune on TRAIN; later VALIDATION selects Ridge narrowly over Elastic Net.                               |
| Runtime consistency         | Exact Notebook 03 moisture and Notebook 04 anomaly artifacts                     | Ordered 16-feature moisture (with causal prior-lab state) and 15-feature process-only anomaly contracts run on TEST replay. |
| Novelty path                | One-Class SVM; validation flag rate 5.925%                                       | 6,000-row fit sample and separate 20,000-row check; display risk is not probability.                                        |
| Database smoke replay       | 1,500 rows; 2 sparse observations; 0 flags / 0 events; latency 9 / 47 ms avg/max | Bounded held-out replay verifies inference, writes and views; not a full-volume deployment test.                            |
| Canonical source            | 1,589,760 process rows; 1,104 sparse targets; 1,103 aligned supervised rows      | One 92-day source feeds all four notebooks and the held-out runtime replay.                                                 |
| Automated Python tests      | 33 passed plus 8 subtests                                                        | Verifies implementation, model artifact, supplied-source hash and safety contracts.                                         |
| Power BI backend validation | Five of five SQL/TMDL physical contracts pass                                    | Live dashboard, contributor, lab, event and overview views are column-compatible and populated as required.                 |

*Source: Notebook, training report, runtime log and validation outputs [11, 12, 10, 15, 16, 17].*

## 8.2 Validation strategy

Validation operates at several levels. Unit and integration-contract tests cover timestamp parsing, selected-grid checks, residence-time alignment, prevention of future-row use, exact ordered feature consistency, artifact inference, deterministic generation and hashes, finite replay defaults, structured database configuration, anomaly mapping, diagnostic rules, temporal patterns, subsystem scoring, severity, feedback and database/runtime helpers. The final run passed 33 tests plus 8 subtests with zero failures [16]. Execution time is deliberately omitted because it varies with machine load and is not a performance acceptance measure.

Integration evidence includes canonical replay contracts and database views. The final live verifier reads physical `sourceColumn` bindings from all five current TMDL tables and compares them with the PostgreSQL views. Dashboard and overview each exposed 1,500 version-filtered rows and the lab view two rows; contributors and events were empty because the bounded v6 interval contained no anomaly flags, which those optional evidence views allow. All required column contracts passed. These checks give confidence that the committed artifact is internally coherent. They do not replace load, usability or failure-recovery tests in an OCP-managed environment.

### 8.3 What the prototype demonstrates

The work demonstrates six feasibility points:

1. a multi-rate stream can preserve sparse quality measurements while supporting inference on the selected near-real-time replay grid;
2. residence-aware process snapshots and strictly previous laboratory inputs can prevent obvious target leakage;
3. model selection and holdout results can be versioned and reproduced;
4. novelty evidence can be converted into cautious, subsystem-oriented guidance;
5. idempotent storage and purpose-built views can support a frequent DirectQuery workload; and
6. one dashboard can organize process, quality, risk and diagnosis without implying autonomous control.

This is a meaningful Industrial Engineering contribution because it organizes the complete production-information loop. The technical value lies not only in the model score but also in data semantics, traceability, latency, operator interpretation and explicit decision authority.

### 8.4 Limitations

The following limitations are material, not footnotes:

- **No live PCS7 connection.** The industrial screens informed process understanding, but CSV replay substitutes for PCS7/historian acquisition.
- **Synthetic final demonstration data.** Relationships, anomaly episodes and metric values may be more regular than plant behaviour.
- **Limited real historical data.** Real grade changes, campaigns, equipment states, calibration events and rare faults are not represented at industrial scale.
- **Simulated lab timing.** The two-hour sparsity mirrors observed availability, but laboratory logistics, delays and re-tests are simplified.
- **Prototype visualization interval only.** The five-second grid and refresh were deliberately selected for simulation; they do not describe native instrumentation sampling. Development-machine timing also excludes plant networks, gateways, concurrent users, service throttling and high availability.
- **Statistical diagnostic limits.** Percentile bands are not approved operating, alarm, safety or trip limits; diagnosis is not causal proof.
- **No closed-loop control.** The prototype neither recommends immediate setpoint changes nor commands any actuator.

Effective sample size remains a central limitation. The canonical stream contains 1,589,760 process rows, but two-hour laboratory cadence provides 1,104 sparse targets and only 1,103 supervised

rows after the initial insufficient-history sample is removed. Label cadence determines supervised information; dense process rows must not be counted as independent quality labels. The single supplied synthetic chronology, despite its TEST  $R^2$  of about 0.825, still requires representative plant campaigns before any accuracy or stability claim can be generalized.

## 8.5 Industrialization roadmap

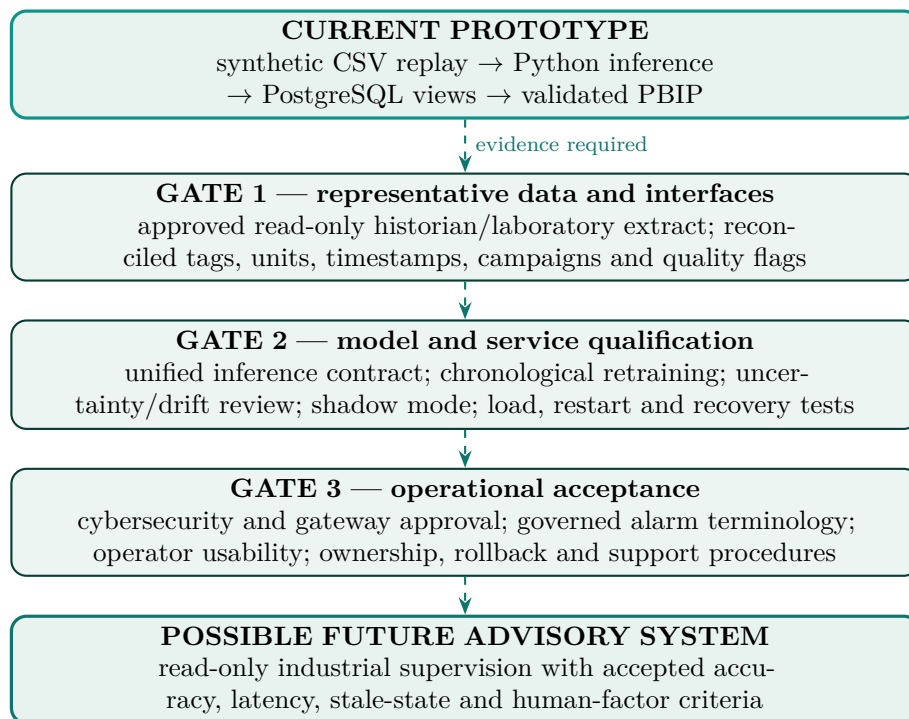
The first future step is data, not control. An approved historian extract should cover normal campaigns, operating modes, grades, maintenance states, instrument quality flags and laboratory timestamps. Variable names, units, sampling behaviour and approved operating envelopes should be reconciled with production and laboratory experts. Models should then be retrained and evaluated by campaign and time period, with uncertainty, drift and false-alarm cost explicitly reviewed.

Second, the existing registry, hashes, finite replay defaults and environment validation should be extended into managed deployment controls: authenticated read-only acquisition; network segmentation; an approved secrets manager; redundant or recoverable PostgreSQL; health checks; retention rules; centralized structured logging; registry-backed promotion and rollback; latency and drift monitoring; backup/restore tests; and Power BI gateway/capacity qualification. A shadow-mode period should compare the solution with PCS7 and laboratory results while it has no operational authority.

Third, diagnosis should improve through representative failure scenarios and operator feedback. Each displayed event can be reviewed as confirmed, rejected or unresolved, with comments and actual cause when known. This can support rule refinement and false-alarm reduction. Alert-management logic should suppress repetitions, respect acknowledged plant states and align with OCP procedures.

Only after real-data validation, safety engineering, cybersecurity approval, operator studies and robust fail-safe design should any recommendation/control architecture be investigated. Even then, a staged advisory or supervisory recommendation layer is more appropriate than immediate autonomous control. OCP operating authority and approved safety systems must remain decisive.

**Recommended next gate.** Run the current software in read-only shadow mode against an approved historian extract, retain laboratory truth unchanged, and conduct joint weekly reviews with production, process, laboratory and automation stakeholders. Progress to operational use only when accuracy, false-alarm burden, timing, failure handling and human factors have documented acceptance criteria.



**Figure 8.1:** Stage-gated path from the current prototype to a possible industrial advisory system.  
*Source: Author's work from the MAP-Dryer-AI repository.*



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# General Conclusion

This internship transformed a concrete production-information delay into a complete Industrial Engineering prototype. Soluble MAP moisture is confirmed by laboratory analysis approximately every two hours while dryer conditions evolve continuously. The response was not to replace the laboratory or control system, but to organize an intermediate decision-support layer: residence-aware moisture prediction, row-level novelty detection, subsystem-oriented diagnosis, traceable PostgreSQL storage and a centralized Power BI supervision interface.

The final implementation is technically coherent. The supplied deterministic 92-day source contains 1,589,760 process rows and 1,104 sparse laboratory observations. It feeds an executable Notebook 01–04 chain, prevents future information from entering the aligned features, and yields 1,103 rows with the original 16 named moisture-model inputs. Direct process values use the residence-adjusted time; density and final product temperature use the most recent strictly previous laboratory result. Notebook 03 selected Ridge using a 772/166/165 chronological TRAIN/VALIDATION/TEST split; TEST MAE is 0.001069, RMSE 0.001403 and  $R^2 = 0.824530$ . Notebook 04 exports the 15-feature anomaly model, scaler and reference profile. The active service loads those exact artifacts while replaying the held-out TEST interval into PostgreSQL and the Power BI views. The five-second interval is a selected prototype replay/visualization interval, not native plant acquisition. The supplied-file hash, artifact hashes, automated tests and live SQL/TMDL validation make these boundaries machine-checkable.

The decisive conclusion remains the scope boundary. The demonstration data are synthetic, the service is not connected to live PCS7, statistical limits are not approved plant limits, and no autonomous action is implemented. Production value can only be assessed after retraining and shadow validation on representative OCP historical data, with operator involvement, industrial cybersecurity, alarm governance and robust operations engineering.

The final classification is therefore exact: this deliverable is an **engineering prototype, not plant-ready**.

From an Operational Excellence perspective, the project shows how data and AI tools can strengthen quality visibility, supervision discipline and problem-solving speed when they are integrated with process understanding and explicit human authority. Its strongest deliverable is therefore not an isolated model, but a transparent architecture and validation path that can support a future OCP industrial study.

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